

**STATISTICAL STUDY OF WORLD COCOA
PRODUCTION AND PRICE**

SUBMITTED BY

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CHAPTER 1

INTRODUCTION

COCOA

Cocoa (*Theobroma cacao* L.) is a perennial tropical crop of immense historical, cultural, and economic significance. Native to the humid lowland rainforests of Central and South America, it has been cultivated for millennia and holds a unique place in human civilization. The earliest known domestication of cocoa is traced back to ancient Mesoamerican societies such as the Olmecs, Mayans, and Aztecs, who considered it a sacred gift from the gods. Archaeological evidence dating as far back as 1500 BCE suggests that cocoa was consumed in the form of a bitter, frothy beverage mixed with chili, spices, or maize. Beyond its nutritional role, cocoa had spiritual, social, and economic functions: it was used in religious offerings, elite social ceremonies, marriage rituals, and even served as a form of currency, underscoring its extraordinary value in pre-Columbian societies. The name *Theobroma*, meaning “food of the gods,” was introduced by Swedish botanist Carl Linnaeus in the 18th century, aptly capturing the reverence this crop has commanded throughout history.

The commercial spread of cocoa beyond the Americas began during the colonial era. Spanish explorers, after encountering cocoa in the 16th century, introduced it to Europe, where its transformation was revolutionary. When sugar, cinnamon, and vanilla were added to the bitter beverage, cocoa became a highly sought-after luxury product among European elites. This surge in demand led to the establishment of extensive cocoa plantations in colonial territories across Africa, Asia, and the Americas, largely dependent on enslaved and indentured labor. Over time, cocoa transitioned from a sacred regional beverage to a global commodity, forming the basis of what we now recognize as the modern chocolate industry.

Economically, cocoa has grown into one of the world’s most traded agricultural products, shaping the livelihoods of millions. Today, it provides income to more than 5 million smallholder farmers, most of whom live in rural, economically vulnerable communities within the “cocoa belt,” located between 20° north and south of the equator. West Africa is the epicenter of global cocoa production: Côte d’Ivoire and Ghana together supply over 60% of the world’s cocoa beans, making the crop vital not only to local economies but also to global trade. Other significant producers

include Indonesia, Nigeria, Ecuador, Cameroon, and Brazil, each contributing to regional supply chains and international markets.

The cocoa value chain is a multi-billion-dollar global industry, heavily driven by the chocolate and confectionery sectors, which account for the majority of demand. Cocoa beans are processed into intermediate products such as cocoa liquor, cocoa butter, and cocoa powder. These are used not only in chocolates but also in beverages, baked goods, cosmetics, pharmaceuticals, and even certain industrial applications. However, the market is subject to volatility, influenced by fluctuating global demand, speculative trading, climate variability, pest and disease outbreaks, and political instability in producing regions. This price instability often directly affects farmer incomes, many of whom already operate on razor-thin margins and remain trapped in cycles of poverty.

In recent decades, issues of sustainability and ethics have become increasingly central to discussions surrounding cocoa. The sector faces multiple challenges: widespread poverty among cocoa farmers, low farm-gate prices, exploitative labor practices including child labor, gender inequities, environmental degradation, and the vulnerability of cocoa cultivation to climate change. Deforestation in West Africa, for instance, has been linked to cocoa expansion, threatening biodiversity and carbon storage. To address these challenges, international initiatives such as Fairtrade certification, Rainforest Alliance programs, and the Cocoa & Forests Initiative have emerged. These aim to promote fair wages, eliminate child labor, encourage gender equality, and incentivize environmentally friendly practices. National governments in Côte d'Ivoire and Ghana have also introduced policies like the Living Income Differential (LID) to ensure farmers receive a minimum price premium, thereby enhancing economic resilience.

Beyond economics and trade, cocoa continues to carry cultural symbolism and global influence. Chocolate has become a universal delicacy, consumed across the world in myriad forms, and it remains strongly tied to celebrations, social traditions, and global cultural identity. At the same time, the crop symbolizes the interwoven histories of colonization, trade, exploitation, and globalization.

Ultimately, cocoa represents much more than the foundation of chocolate. It is a crop that embodies ancient heritage, economic interdependence, and pressing contemporary issues in global agriculture. Its future sustainability depends on balancing rising global demand with responsible agricultural practices, equitable

economic returns for farmers, and the protection of fragile tropical ecosystems. By ensuring that cocoa production is both environmentally sustainable and socially just, the crop can continue to thrive as both a cultural legacy and a driver of livelihoods for generations to come.

Objectives of the Study

- 1.To study the trends in cocoa area, production, and productivity at regional and global levels.
- 2.To analyze domestic and international price behavior and volatility of cocoa.
- 3.To examine the relationship between area, production, productivity, and price using statistical methods.
- 4.To evaluate export and import trends of cocoa and India's role in global trade.
- 5.To forecast future production and price movements through time series analysis.
6. To examine the long-run equilibrium relationship between domestic and international cocoa prices through cointegration analysis.

CHAPTER 2

REVIEW OF LITERATURE

1. Anila, Kuruvila, and Minimol (2024) examined the trend in cocoa cultivation and price behaviour in Kerala from 1978–2019. Their analysis revealed that while the area under cocoa declined over time, productivity steadily improved, resulting in higher overall production. They attributed this to government expansion schemes and research support, particularly after the 1990s. In addition, their time-series analysis of cocoa prices across four major Kerala markets between 2005–2022 showed an increasing trend, with clear seasonal variations. Prices peaked during April–May and dropped to their lowest between August and October, whereas cyclical and irregular fluctuations were insignificant. These findings underscore the influence of seasonal factors on cocoa markets in Kerala and highlight the need for better market intelligence systems to support farmers.

2. George and Raju (2024) analyzed cocoa market performance in India using econometric models to examine supply responses and price variability. Their study emphasized that India, though a small player globally, has shown consistent growth in cocoa production due to intercropping practices and government initiatives. The authors highlighted that farmers' allocation decisions are influenced not only by price signals but also by climatic conditions, policy support, and availability of high-yielding varieties. Employing advanced modeling, they observed that cocoa markets are prone to volatility, and both lagged production and price factors significantly shape supply trends. Their findings align with earlier works such as Patil et al. (2012) and Jaweriah et al. (2020), confirming that price variability alone does not fully explain acreage response. Instead, a combination of economic and non-economic factors drives farmers' choices.

3. Sundariya, Banumathy, and Ravichandran (2022) examined cocoa growth and supply response in Tamil Nadu using CAGR and a Nerlovian adjustment model. Their findings indicated significant positive growth in area and production in Coimbatore district, while Thanjavur experienced a decline due to climatic disturbances. The study highlighted the importance of lagged area and price in influencing acreage response, with production decisions largely shaped by past outputs. Price variability showed

mixed effects across districts, with farmers requiring two to three years to adjust cultivation patterns. The authors suggested policy interventions such as transport subsidies, procurement centres, crop insurance, and promotion of high-yielding varieties to stabilize production and prices. Their work emphasizes how both price and non-price factors significantly shape cocoa supply responses in India.

4. Gilbert (2012) analyzed the dynamics of world cocoa prices using a structural econometric model spanning 1950–2012. The study revealed that supply shocks, such as crop shortfalls, cause prolonged price increases by reducing consumption and depleting inventories. Demand shocks, particularly fluctuations in grindings, were also found to exert equally significant and lasting effects, challenging the assumption that supply factors dominate cocoa price behavior. The research further identified weak but notable interlinkages with the coffee market and emphasized the partial role of storage in buffering shocks. Overall, the study highlights the complex interplay of supply, demand, and macroeconomic forces in cocoa price determination.

5. Adams et al. (2023) investigate stakeholder perceptions of the Living Income Differential (LID) policy introduced by major West African cocoa producers to improve farmer livelihoods. Drawing on interviews with industry actors, farmer representatives, policymakers, and NGOs, the study explores the policy's intended social benefits and unintended market impacts. Findings reveal mixed views: while many stakeholders see the LID as a positive step toward addressing income inequality, concerns arise over buyer resistance, adjustments in origin differentials, and potential shifts in sourcing. The authors emphasize the importance of complementary measures, transparency, and collaborative governance to ensure the LID's effectiveness in achieving sustainable incomes.

6. Christopher L. Gilbert (2015) extends the historical cocoa price series back to the late 19th century, enabling a long-run analysis of multi-decade price cycles. He links these cycles to the biological characteristics of cocoa production particularly long tree lifespans and delayed supply responses combined with periods of demand expansion and stagnation. Gilbert identifies recurrent boom–bust patterns, where high prices stimulate planting booms, followed by oversupply and prolonged price declines. He also examines how structural changes, such as the rise of West Africa as the dominant

supplier, have influenced volatility and price trends. The chapter provides historical context for modern cocoa market instability.

7.Francis Ruf (1995) examines how futures market speculation influences short-run cocoa price movements, interacting with underlying supply demand fundamentals. He argues that while weather shocks, political events, and crop cycles set the stage for price changes, speculative trading can amplify these shifts, contributing to overshooting and heightened volatility. Ruf highlights the feedback loop between speculative expectations and physical market perceptions, noting that price spikes or slumps can occur even without immediate changes in fundamentals. The chapter underscores the importance of understanding speculative dynamics for interpreting price behavior and designing stabilization policies in globally traded agricultural commodities like cocoa.

8.Marcella Vigneri and coauthors (2018) investigate how farm-gate pricing institutions have contributed to Ghana's cocoa sector recovery. They show that maintaining relatively high and stable producer prices set through the state-led marketing system helped restore farmer confidence, incentivize farm rehabilitation, and encourage adoption of yield enhancing inputs. The policy's predictability reduced income uncertainty, enabling longer-term investment decisions. The authors also find evidence of productivity gains stemming from complementary public programs, such as input subsidies and extension services. However, they caution that continued reliance on high prices requires fiscal sustainability and resilience against global price downturns to protect both farmers and public budgets.

9.Ruf (2019) examines the Living Income Differential (LID) in the context of persistent farmer poverty traps in Côte d'Ivoire and Ghana. The study argues that while the LID raises cocoa prices and provides short-term income support, it does not fully resolve structural barriers that keep farmers in poverty, such as limited access to credit, low productivity, and dependency on a single cash crop. Ruf emphasizes that without complementary measures—like agricultural extension services, diversification, and improved governance—higher cocoa prices alone cannot guarantee sustainable livelihoods. The paper positions LID as a partial solution, highlighting the need for integrated policy interventions.

CHAPTER 3

METIRIALS AND METHODS

Data Source

1. Directorate of Cashew and Cocoa Development (DCCD):

The DCCD provides comprehensive statistical data on cocoa cultivation in India, including area, production, and productivity. It also offers international data on major cocoa-growing countries, covering domestic and international prices, as well as export and import details of cocoa and cocoa products.

2. Food and Agriculture Organization Statistics (FAOSTAT):

FAOSTAT provides comprehensive data on global cocoa area, production, and yield. These statistics were utilized to examine long-term trends and growth patterns in cocoa cultivation and productivity at both national and international levels.

3. Directorate of Economics & Statistics (DES):

The Directorate of Economics & Statistics (DES) provides official statistical data on cocoa cultivation in India, including area, production, productivity, market prices, and trade-related information, which are widely used for agricultural and economic analysis.

3.1 Descriptive Statistics

Descriptive statistics involves methods and techniques used to summarize, organize and present data in a meaningful way. It provides a clear and concise summary of the main characteristics of a datasets such as central tendency (mean, median, mode) and dispersion (range, variance and standard deviation). Descriptive statistics are fundamental in both research and everyday applications offering a structured approach to understanding the quantitative aspects of information

3.2 Correlation Analysis

Correlation analysis is a statistical technique used to examine the strength and direction of the linear relationship between two quantitative variables. This method provides insights into how one variable may change in relation to another. The primary metric used in correlation analysis is the correlation coefficient, denoted as r . This coefficient ranges from -1 to +1. A value of +1 indicates a perfect positive correlation,

meaning that as one variable increases, the other variable also increases proportionally. Conversely, a value of -1 signifies a perfect negative correlation, where one variable decreases as the other increases. A value of 0 indicates no linear correlation between the variables.

There are several different measures for the degree of correlation in data, depending on the kind of data: principally whether the data is a measurement, ordinal, or categorical.

3.2.1 Pearson Correlation

The Pearson product-moment correlation coefficient, also known as r , R , or Pearson's r , is a measure of the strength and direction of the linear relationship between two variables that is defined as the covariance of the variables divided by the product of their standard deviations. This is the best-known and most commonly used type of correlation coefficient. When the term "correlation coefficient" is used without further qualification, it usually refers to the Pearson product-moment correlation coefficient.

3.3 Time Series Analysis

A time series is a set of observations Y_t , each one being recorded at a specific time t . The main objective of the times series analyses is forecasting. In Forecasting, we can estimate the future values of the series. Good forecasts enable the analyst to take action so as to control a given process. Time Series arise as recordings of processes that vary over time. A recording can be either a continuous trace or a set of discrete observations. Time series data occur naturally in many application areas such as economics, finance, environmental, medicine etc.

What sets time series data apart from other data is that the analysis can show how variable change over time. In other words, time is a crucial variable because it shows how the data adjusts over the course of the data points as well as the final results. It provides an additional source of information and a set order of dependencies between the data. The time series analysis typically requires a large number of data points to ensure consistency and reliability. An extensive data set ensures you have a representative sample size and that analysis can cut through noisy data. It also ensures that any trends or patterns discovered are not outliers and can account for seasonal variance. Additionally, time series data can be used for forecasting -predicting future data based on historical Data.

Classical times series analysis, generally known as Box and Jenkins time series approach, deals with the modelling and analysis of finite variance linear time series models. Classical times series models are a family of models that have been traditionally used a lot in many domains for forecasting. They are strongly based on temporary variation inside a time series and they work well with univariate time series. Some advanced options exist to add external variables into the models as well.

The first step in a time series analysis is to draw a time series plot. A time series plot is a plot that plots all the observations against time. It should include all the important features of data such as trend, seasonality, outliers, smooth changes in the structure, turning points, or sudden discontinuity which plays a vital role in describing the data and also to formulate a sensible model and in choosing an appropriate forecasting method.

Components For Time Series Analysis

The various reasons or the forces which affect the values of an observation in a time series are the components of a time series. The four categories of the components of time series are

1. Trend
2. Seasonal Variations
3. Cyclic Variations
4. Random or Irregular movements

Seasonal and Cyclic Variations are the periodic changes or short-term fluctuations.

Trend

This is the long- term upward or downward movement of the series due to factors that affect the mean of the series. Some series may remain more or less at a constant level. Sudden or frequent changes are incompatible with the idea of trend. Thus, trend may be defined as a slowly changing non-random component of a time series. A trend is general, smooth, long-term, and average tendency. It is observable that the tendencies may increase, decrease or be stable in different sections of time. But the overall trend must be upward, downward, or stable. The population, agricultural production, items manufactured, number of births and deaths, number of industries or any factory, and number of schools or colleges are some of its examples showing some kind of tendencies of movement.

Seasonal Variations

Seasonal variations are due to the rhythmic forces which operate in a regular and periodic manner over a span of less than a year. They have the same or almost the same pattern year after year. Many time series in business and industry show variations in values from one season to the other, which repeat quarterly, monthly, and annually. In time series data where only the annual figures are given, there are no seasonal variations. There are seasonal variations which are resulted from natural forces like weather conditions and resulted from man-made conventions like habits, fashion, etc. The main objective of the measurement of seasonal variations is to isolate them from trends and study their effects.

Cyclic Variations

Cyclic variations are the oscillatory movements in a time series with a period of oscillation being more than one year. Once complete period is called a cycle. They are regular variations but not periodic, and normally last from seven to eleven years. Moreover, the average length of a cycle is usually longer than that of seasonality and the magnitude of a cycle is usually more variable than that of seasonality.

The major distinction between a seasonal and a cyclical pattern is that the former is of a constant length and recurs on a regular periodic basis, while the latter varies in length. Moreover, the average length of a cycle is usually longer than that of seasonality.

Irregular Or Random Components

Apart from the regular variations, almost all the series contain another factor called the random or irregular components which are accounted for the secular trend, seasonal and cyclic variations. These fluctuations are purely random, erratic, unforeseen, and unpredictable. It is not possible to isolate irregular variations, nor one can forecast or estimate them precisely. It is also known as residual fluctuations.

Linear and Non-Linear Trend

If we plot the time series values on a graph in accordance with time t . The pattern of the data clustering shows the type of trend. If the set of data clusters more or less around a straight line, then the trend is linear; otherwise, it is non-linear (Curvilinear).

Mathematical Models For A Time Series

The following are the two models commonly used for the decomposition of a time series into components, they are

- i. Decomposition by additive model

ii. Decomposition by multiplicative model

i. Decomposition By Additive Model

In an additive model, a time series can be expressed as

$$Y_t = T_t + S_t + C_t + R_t$$

Where Y_t is the time series at time t , T_t is the trend value, and S_t , C_t and R_t represents the seasonal, cyclic, and random components at time t . The term S_t does not appear in the series of annual data. In an additive model, it will have positive or negative values and in long run, $\sum R_t$ will be zero. The additive model assumes that all four components of the time series operate independently of each other so that none of these components has any effect on the remaining components.

ii. Decomposition By Multiplicative Model

In a multiplicative model, time series can be expressed as

$$Y_t = T_t \times S_t \times C_t \times R_t$$

Where Y_t is the time series at time t , T_t is the trend value, and S_t , C_t and R_t represents the seasonal, cyclic, and random components at time t . According to the multiplicative model, R_t instead of assuming positive or negative values fluctuates above or below unity and the geometric mean of R_t , in the long run, is unity.

It is noted that the multiplicative model can be converted into an additive model by taking logarithms. Since,

$$\log Y_t = \log T_t + \log S_t + \log C_t + \log R_t$$

Hence it is noted that the multiplicative model $R_t > 0$. Hence $\log R_t$ is positive or negative according to $0 < R_t < 1$ or $R_t > 1$, which is desirable.

The assumption underlying the additive model is that there is no interaction among the different components and the multiplicative model assumes the presence of such interactions. So that the suitability of one scheme compared to the other depends on the nature of the time series. For instance, the multiplicative model may be more appropriate in the study of the time series on price, if it is the case that in an inflationary situation, the seasonal component is reinforced by the rising trend in

prices. Whereas, in a deflationary situation the falling trend in price dampens these components.

Determination Of Trend Component

The main objectives of the measurement of trend are to eliminate the trend from the series and to study the trend itself or to forecast the future behaviour of the trend. The trend can be studied and measured by the following methods:

- a) Free hand curve fitting method
- b) Method of semi- averages
- c) Method of curve fitting by principle of least squares
- d) Method of moving averages

Compound Annual Growth Rate (CAGR)

To estimate the compound annual growth rate (CAGR) of Cocoa in terms of area, production, and productivity, a semi-logarithmic trend equation was employed. The functional form of the model is expressed as:

$$\ln Y_t = \alpha + \beta t + u_t$$

where Y_t represents the variable under study (area, production, or productivity), t denotes the time period taking values 1, 2, 3, ..., n for successive years, α is the intercept term, β is the regression coefficient that measures the rate of change over time, and u_t is the random error term. The logarithmic transformation of Y_t helps to linearize the exponential growth pattern commonly observed in agricultural time series data, making estimation through ordinary least squares (OLS) more reliable.

Once the regression equation is estimated, the compound annual growth rate (CAGR) is derived from the slope coefficient (β) using the following relationship:

$$\text{CAGR} = [(\text{Antilog of } \beta) - 1] \times 100$$

This formulation converts the natural-logarithmic growth coefficient into a percentage rate, representing the average annual change in the variable over the study period. A positive CAGR indicates a consistent upward trend, while a negative value reflects a declining tendency. This method is widely applied in agricultural economics to analyze long-term growth performance and temporal stability across different production parameters.

Cuddy–Della Valle Instability Index (CDVI)

Instability in area, production, and productivity was measured using the Cuddy–Della Valle Index (1978), which corrects the overestimation bias of the simple Coefficient of Variation (CV) by adjusting for the trend component. The index is expressed as:

$$I = CV \times \sqrt{(1 - R^2)}$$

Where:

- I = Instability Index (in percent)
- CV = Coefficient of Variation (in percent)
- R^2 = Adjusted Coefficient of Determination from the trend regression

The classification of instability is as follows:

- Low instability: 0–15%
- Medium instability: 15–30%
- High instability: >30%

This index provides a more reliable measure of instability in time-series data by accounting for the influence of trends.

Augmented Dickey-Fuller (ADF) Test

The Augmented Dickey–Fuller (ADF) test is a widely used statistical test for checking the presence of a unit root in a time series, which helps determine whether the series is stationary or non-stationary. Stationarity is an essential assumption in many time series models because non-stationary data may lead to spurious regression results. The ADF test extends the basic Dickey–Fuller test by including lagged differences of the dependent variable to account for higher-order serial correlation. The general form of the ADF regression is:

$$\Delta Y_t = \alpha + \beta t + \gamma Y_{t-1} + \sum \delta_i \Delta Y_{t-i} + \epsilon_t$$

where:

- $\Delta Y_t = Y_t - Y_{t-1}$ \Delta Y_t = Y_t - Y_{t-1} is the first difference of the series,
- α is a constant (drift term),
- βt represents the deterministic time trend,
- γ is the coefficient that tests for the presence of a unit root,

- δ_i are coefficients of the lagged differenced terms to remove autocorrelation, and
- ϵ_t is the white noise error term.

The null hypothesis $H_0: \gamma = 0$ indicates the presence of a unit root (non-stationary series), while the alternative hypothesis $H_1: \gamma < 0$ implies stationarity.

Engle-Granger Pairwise Cointegration Test

The Engle-Granger Pairwise Cointegration Test is a two-step procedure used to determine whether two non-stationary time series share a long-run equilibrium relationship. If a linear combination of two individually non-stationary series is found to be stationary, then the variables are said to be cointegrated. This method is commonly applied in econometrics when testing for long-run relationships between economic or financial variables.

Step 1 – Estimation of Long-Run Relationship:

A static regression is estimated between the two variables:

$$Y_t = \alpha + \beta X_t + u_t$$

where

- Y_t is the dependent variable
- X_t is the independent variable
- α and β are parameters
- u_t represents the residuals.

Step 2 – Residual Stationarity Test:

The residuals u_t obtained from the regression are tested for stationarity using the Augmented Dickey-Fuller (ADF) test:

$$\Delta u_t = \rho u_{t-1} + \sum \delta_i \Delta u_{t-i} + \epsilon_t$$

The null hypothesis $H_0: \rho = 0$ indicates that the residuals contain a unit root (no cointegration), while the alternative hypothesis $H_1: \rho < 0$

$0 < \rho < 1$ suggests that the residuals are stationary, meaning the two variables are cointegrated.

Thus, if the residuals are stationary, the test concludes that a long-run equilibrium relationship exists between the two non-stationary time series.

Johansen Eigenvectors

The Johansen Cointegration Test is a multivariate statistical method used to test the existence and number of cointegrating relationships among two or more non-stationary time series. Unlike the Engle–Granger test, which is limited to a single cointegration vector, the Johansen approach allows for the identification of multiple cointegration relationships within a system of variables, making it particularly suitable for vector autoregressive (VAR) models.

The starting point is a Vector Autoregressive (VAR) model of order p :

$$Y_t = \mu + \Phi_1 Y_{t-1} + \Phi_2 Y_{t-2} + \dots + \Phi_p Y_{t-p} + \varepsilon_t$$

This can be re-parameterized into a Vector Error Correction Model (VECM) form:

$$\Delta Y_t = \mu + \Pi Y_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta Y_{t-i} + \varepsilon_t$$

where:

- Y_t is a vector of n non-stationary time series,
- ΔY_t denotes the first differences,
- μ is a constant term,
- Γ_i are short-run adjustment coefficients,
- $\Pi = \sum \Phi_i - I$ contains information about the long-run relationships,
- ε_t is the white noise error.

The key lies in the rank of the matrix Π :

- If $\text{rank}(\Pi) = 0 \rightarrow$ no cointegration.
- If $\text{rank}(\Pi) = r$ ($0 < r < n$) $\rightarrow$ there are r cointegration vectors.

- If $\text{rank}(\Pi) = n \rightarrow$ all variables are stationary.

Johansen proposes two likelihood ratio tests to determine the number of cointegrating vectors (r):

1. Trace Statistic:

$$\lambda_{\text{trace}}(r) = -T \sum \ln(1 - \lambda^i)$$

2. Maximum Eigenvalue Statistic:

$$\lambda_{\text{max}}(r, r+1) = -T \ln(1 - \lambda^{r+1})$$

where:

- T is the sample size,
- λ^i are the estimated eigenvalues.

The null hypothesis is that there are at most r cointegration vectors, against the alternative that there are more.

CHAPTER 4

ANALYSIS OF DATA

KERALA AREA, PRODUCTION AND PRODUCTIVITY (1980-2024)

Table 4.1 Average Cocoa Area, Production, and Productivity in Kerala (1980–2024)

TIME INTERWEL	AVERAGE AREA(HA)	AVERAGE PRODUCTION(MT)	AVERAGE PRODUCTIVITY(KG/HA)
1980-1989	1831.4	576.3	285.5
1990-1999	2039.6	1121.4	568.9
2000-2009	5273.2	9028.2	548.8
2010-2019	6746.9	14025	486.7
2020-2024	10049	117783.8	564.6

SOURCE: DIRECTORATE OF CASHEWNUT AND COCOA DEVELOPMENT (DCCD) AND DIRECTORATE OF ECONOMICS & STATISTICS (DES)

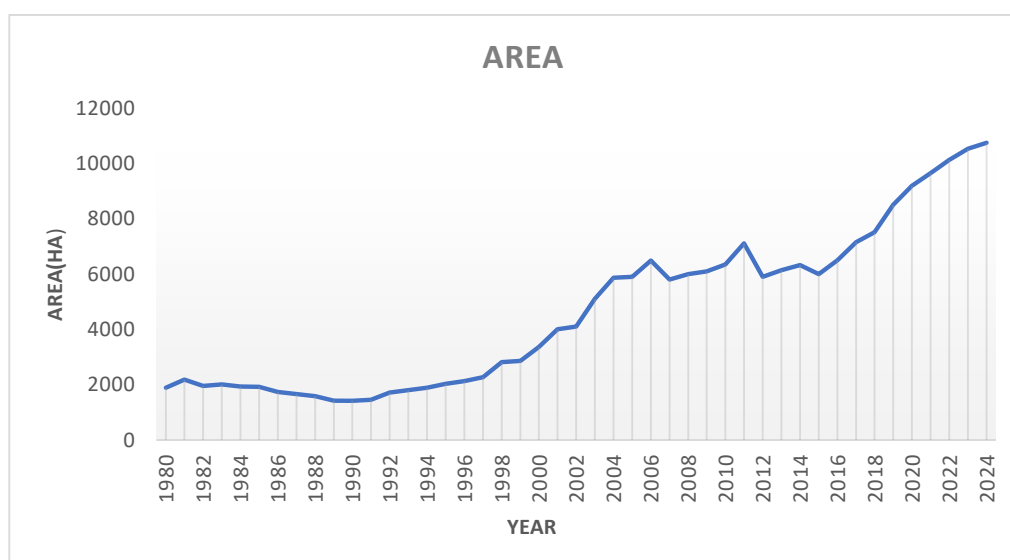


FIG 4.1 KERALA COCOA AREA TRENDS FROM 1980 TO 2024

The trend in cocoa cultivated area in Kerala from the early 1980s to the early 2020s shows a long-term expansion with short-term fluctuations, reflecting changing economic incentives and farming strategies. In the 1980s, cocoa area remained relatively stable around 1,800–2,000 ha, followed by a gradual decline toward the late 1980s and early 1990s, indicating limited farmer interest and lower profitability compared to competing plantation crops.

From the mid-1990s, cocoa area began to rise steadily as farmers increasingly adopted cocoa as an intercrop in coconut and arecanut gardens. This expansion accelerated

during the early 2000s, with area increasing sharply up to around 2005–06, driven by favorable prices, government support, and growing demand from the domestic chocolate industry. A brief contraction around 2007–08 reflects market and management adjustments, but the overall upward trend resumed thereafter.

During 2010–2015, cocoa area showed moderate fluctuations, suggesting a phase of consolidation rather than rapid expansion. From 2016 onwards, the area increased sharply again, reaching its highest level by 2023–24, indicating renewed farmer interest and confidence in cocoa as a remunerative intercrop. Overall, the trend highlights that cocoa expansion in Kerala has been market-driven and intercropping-based, and future growth will depend less on new area expansion and more on sustaining productivity and plantation health.

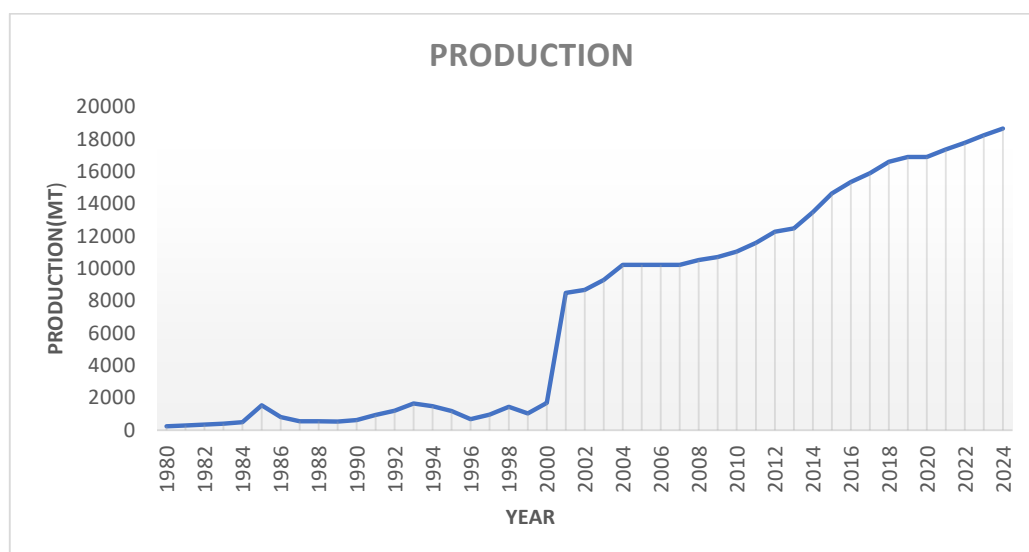


FIG 4.2 KERALA COCOA PRODUCTION TRENDS FROM 1980 TO 2024

The trend in cocoa production in Kerala shows a clear transition from a marginal crop to a commercially significant intercrop over the study period. During the 1980s and early 1990s, cocoa production remained very low and fluctuating, reflecting limited cultivation, young plantations, and low yield levels. Production gains during this phase were modest and unstable, indicating the experimental nature of cocoa adoption in the state.

A major structural break occurs around 2000–01, when cocoa production rises sharply, corresponding with the maturation of plantations established in the 1990s and rapid expansion of cocoa area as an intercrop in coconut and arecanut systems. From 2002

onwards, production crossed 10,000 tonnes and continued to increase steadily, reflecting both area expansion and improvements in management practices.

Between 2005 and 2012, production showed gradual growth with minor fluctuations, suggesting consolidation of the sector. From 2013 onwards, cocoa production increased sharply and consistently, reaching its highest level by 2023–24, supported by mature plantations, favorable price trends, and rising domestic demand from the chocolate and confectionery industry.

Overall, the production trend indicates that Kerala's cocoa growth has been structurally driven by area expansion and plantation maturity, with recent increases suggesting improved stability. However, sustaining this upward trajectory will increasingly depend on yield enhancement, disease management, and rejuvenation of aging plantations, rather than further expansion of cultivated area.

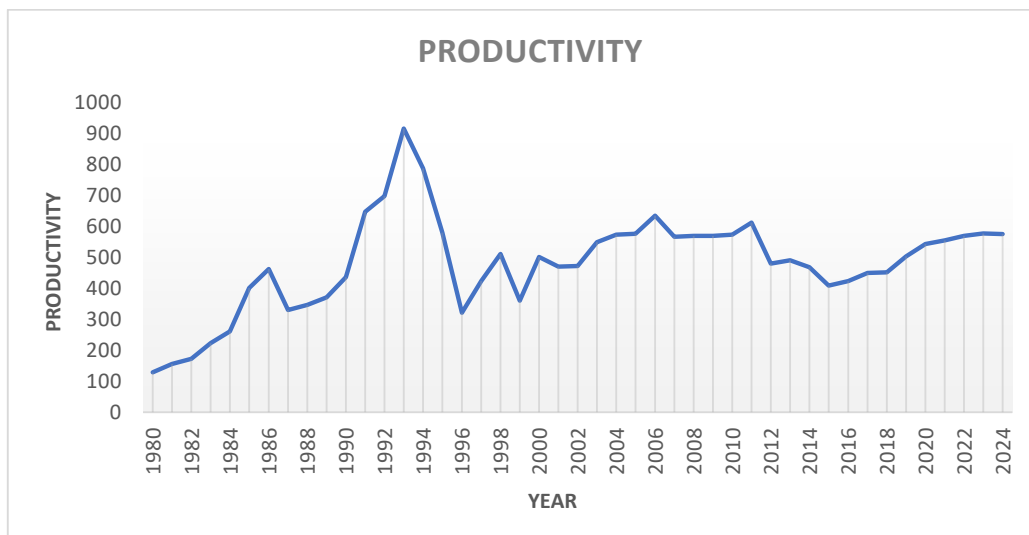


FIG 4.3 KERALA COCOA PRODUCTIVITY TRENDS FROM 1980 TO 2024

The trend in cocoa productivity in Kerala reveals pronounced fluctuations with no sustained long-term upward trajectory, highlighting the dominance of management, climatic, and biological factors. During the early 1980s, productivity was relatively low but increased steadily through the mid-1980s, reflecting gradual adoption of better cultivation practices. A sharp rise in the early 1990s, culminating in a peak around 1992–93, indicates a period of highly favorable conditions, likely supported by young, vigorous plantations and improved husbandry.

This peak was followed by a steep decline in the mid-1990s, suggesting the onset of yield stress due to aging trees, pest and disease incidence, and inconsistent farm management. From the late 1990s to the mid-2000s, productivity recovered and stabilized at a moderate level (around 500–600 kg/ha), indicating a balance between plantation maturity and management practices.

During 2008–2014, productivity again declined and showed considerable variability, reflecting climatic stress, rising labor costs, and reduced input use. However, from 2015 onwards, productivity exhibited a gradual recovery, reaching relatively stable levels by 2022–24, likely due to better price incentives, selective replanting, and improved crop management.

Overall, the productivity trend suggests that Kerala’s cocoa sector has relied more on area expansion than sustained yield improvement, emphasizing the need for technological interventions, replanting of senile trees, and consistent extension support to achieve stable and long-term productivity growth.

COMPOUND ANNUAL GROWTH RATE (CAGR)

Table 4.2 Compound Annual Growth Rates of Area, Production, and Productivity of Cocoa in Kerala

TIME INTERWEL	AREA(HA)	PRODUCTION(MT)	PRODUCTIVITY(KG/HA)
1980-1999	1.18	7.55	6.47
2000-2024	3.72	5.17	-0.11
TOTAL	4.92	11.25	1.62

SOURCE: DIRECTORATE OF CASHEWNUT AND COCOA DEVELOPMENT (DCCD) AND DIRECTORATE OF ECONOMICS & STATISTICS (DES)

The compound annual growth rate (CAGR) of cocoa cultivation in Kerala across the two time intervals (1980–1999 and 2000–2024) and the total period reveals distinct patterns of development in area, production, and productivity. During 1980–1999, cocoa cultivation in Kerala grew moderately in area at 1.18%, but production increased sharply by 7.55%, while productivity rose impressively by 6.47% per annum. This period marks the initial development phase of cocoa in the state, driven by the introduction of cocoa as an intercrop, improved planting materials, and strong institutional support from the Directorate of Cashew and Cocoa Development (DCCD). The rapid rise in productivity indicates effective agronomic management, better extension services, and favorable climatic conditions that enhanced yield efficiency.

In contrast, the period 2000–2024 shows a different trend area expansion accelerated to 3.72%, but production growth slowed to 5.17%, and productivity slightly declined by –0.11%. This reflects a shift toward area-led growth rather than yield-led improvement. The decline in productivity can be linked to aging plantations, pest and disease incidence, limited replanting, and uneven adoption of scientific practices. While the increase in area and production demonstrates sustained interest among farmers and support through government schemes, stagnating yields highlight challenges in technological dissemination and farm management.

Considering the total period (1980–2024), cocoa area in Kerala expanded at 4.92%, and production recorded a strong growth of 11.25%, while productivity grew modestly by 1.62%. Overall, this pattern indicates that Kerala’s cocoa sector has made remarkable progress in terms of expansion and output, though future growth will depend on addressing productivity bottlenecks through replanting programs, improved hybrids, pest management, and enhanced extension services to achieve sustainable and efficient production.

INDIA COCOA AREA ,PRODUCTIONAND YIELD TRENDS FROM 1977 TO 2023

Table 4.3 Average India Cocoa Area, Production, and Yield (1961–2023)

TIME INTERWEL	AREA(HA)	PRODUCTION(MT)	YIELD(KG/HA)
1977-1986	14201	4360	448.32
1987-1996	17439	6179.6	378.88
1997-2006	18922.1	7290.9	397.46
2007-2016	55807.5	13386	257.69
2017-2023	97428.57143	24857.14286	253.6714

SOURCE: FAOSTAT (FOOD AND AGRICULTURE ORGANIZATION OF THE UNITED NATIONS).

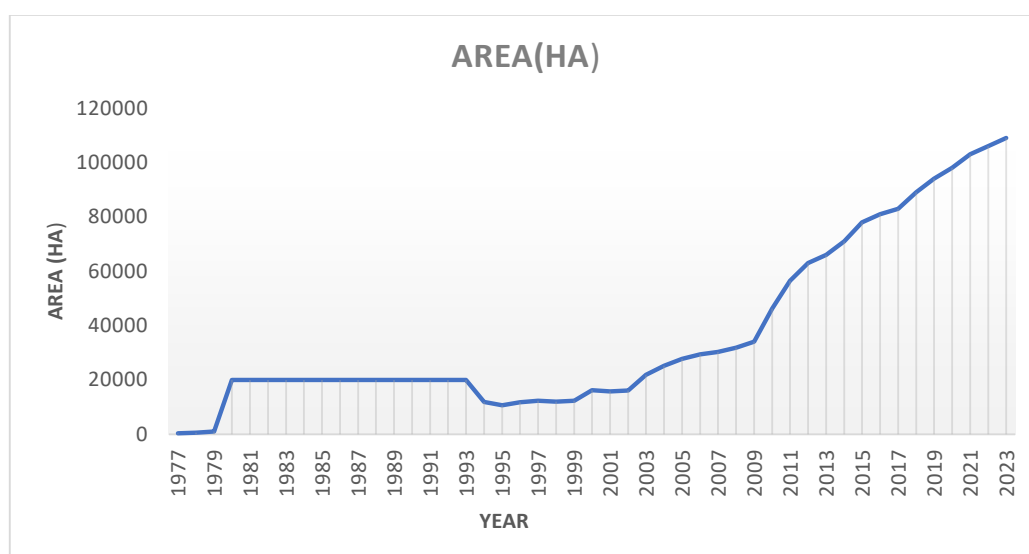


FIG 4.4 INDIA COCOA AREA TRENDS FROM 1961 TO 2023

The graph illustrates the trend in India's cocoa cultivation area (in hectares) from 1977 to 2023. In the initial years, the area under cocoa was very limited, but it increased sharply by the early 1980s and then remained nearly stable around 20,000 hectares for over a decade. During the 1990s, a decline was observed, reflecting reduced farmer interest due to low profitability, inadequate market infrastructure, and limited government focus on the crop. However, from the early 2000s onward—especially after 2005—the cocoa area in India began to expand steadily, showing a remarkable surge after 2010. This rapid growth corresponds with government initiatives promoting cocoa as an intercrop with coconut and arecanut, rising domestic demand from the chocolate industry, and improved availability of high-yielding hybrid varieties. By 2023, the cocoa area surpassed 100,000 hectares, indicating India's

increasing emphasis on diversifying agricultural income sources and strengthening its position in the global cocoa sector.

INDIA COCOA PRODUCTION TRENDS FROM 1977 TO 2023

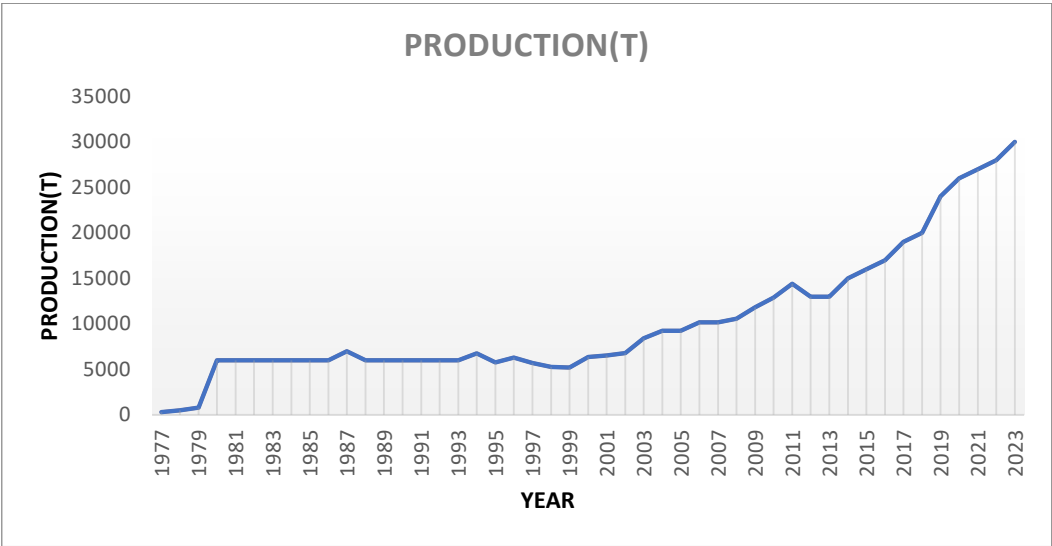


FIG 4.5 INDIA COCOA PRODUCTION TRENDS FROM 1961 TO 2023

The graph represents the trend in India’s cocoa production (in tonnes) from 1977 to 2023. During the late 1970s, production was very low, but it increased sharply by the early 1980s as cocoa cultivation expanded in southern states, particularly Kerala, Tamil Nadu, Karnataka, and Andhra Pradesh, where it was promoted as an intercrop with coconut and arecanut. From the mid-1980s to the late 1990s, production remained relatively stagnant around 6,000 tonnes, mainly due to limited technological advancement, low productivity, and poor farmer awareness. However, after 2000, cocoa production began to rise steadily, reflecting government and private sector initiatives, including support from the Directorate of Cashew and Cocoa Development (DCCD) and growing collaboration with chocolate manufacturing industries. The sharp upward trend after 2010 indicates the positive impact of improved planting materials, scientific cultivation methods, and increased domestic demand for cocoa-based products. By 2023, India’s cocoa production surpassed 30,000 tonnes, highlighting the crop’s growing economic importance and the country’s progress toward self-reliance in cocoa production.

INDIA COCOA YIELD TRENDS FROM 1977 TO 2023

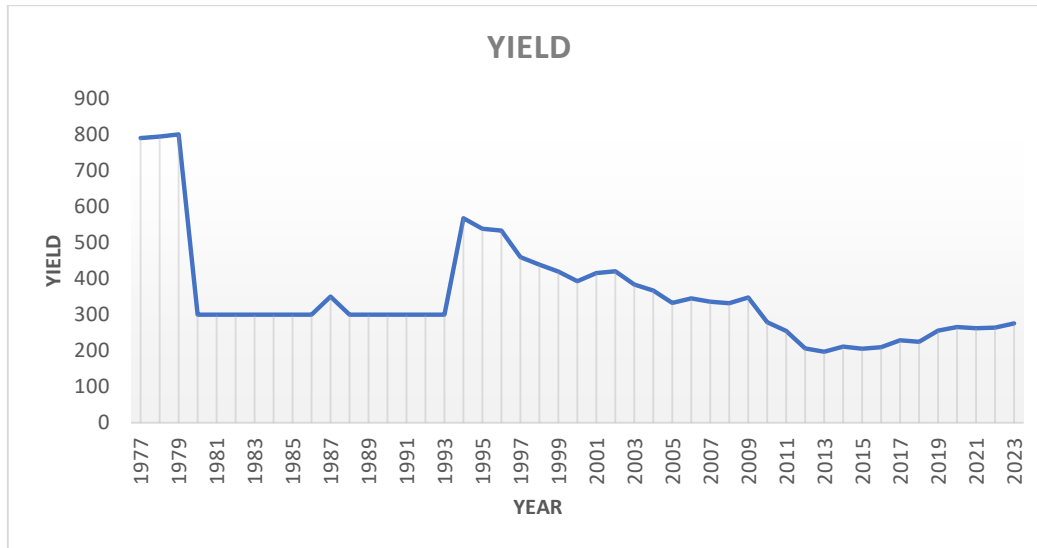


FIG 4.6 INDIA COCOA YIELD TRENDS FROM 1961 TO 2023

The graph illustrates the trend in India's cocoa yield (kg/ha) from 1977 to 2023. Initially, yield levels were high, around 800 kg/ha in the late 1970s, reflecting good productivity from limited, well-managed plantations. However, a sharp decline occurred in the early 1980s, stabilizing around 300 kg/ha for several years due to expansion into new areas with less suitable agro-climatic conditions and inadequate technical support. A temporary improvement during the early 1990s, when yield touched nearly 600 kg/ha, was likely the result of better management practices and the introduction of improved varieties. Yet, from the late 1990s onward, yields gradually declined, reaching their lowest levels around 2010–2013, due to aging plantations, poor maintenance, insufficient nutrient management, and pest or disease pressures. In recent years, a slight upward trend is visible after 2015, indicating the effects of replanting with high-yielding hybrids, improved extension services, and greater farmer awareness. Overall, while India's cocoa yield has shown fluctuations, the recent recovery suggests gradual improvement driven by modernization and scientific interventions in cocoa cultivation.

COMPOUND ANNUAL GROWTH RATE (CAGR)

Table 4.4 Compound Annual Growth Rates of Area, Production, and Yield of Cocoa in India

	AREA(HA)	PRODUCTION(MT)	YIELD(KG/HA)
1977-2000	7.49	6.98	-0.47
2001-2023	9.55	6.8	-2.51
TOTAL	7.05	5.54	-1.4

SOURCE: FAOSTAT (FOOD AND AGRICULTURE ORGANIZATION OF THE UNITED NATIONS).

The compound annual growth rate (CAGR) analysis of India's cocoa sector reveals a clear contrast between the two sub-periods and the overall trend. During 1977–2000, cocoa area expanded rapidly at 7.49% per annum, while production grew at a comparable rate of 6.98%, indicating that output growth was largely supported by land expansion, though the slight decline in yield (-0.47%) suggests early signs of productivity constraints. In the subsequent period, 2001-2023, the growth in area further accelerated to 9.55%, reflecting increased adoption of cocoa as an intercrop and policy-driven expansion; however, production growth slowed to 6.80%, accompanied by a sharp deterioration in yield (-2.51%), pointing to growing inefficiencies due to aging plantations, suboptimal crop management, pest and disease pressures, and climatic stress. The overall period (1977-2023) shows that while cocoa area grew strongly at 7.05%, production increased at a lower rate of 5.54%, resulting in a significant negative yield growth (-1.40%). This divergence clearly indicates that India's cocoa growth has been predominantly area-led and not productivity-driven, underscoring the urgent need for yield-enhancing technologies, improved planting material, better extension services, and climate-resilient production practices to ensure sustainable long-term growth of the cocoa sector.

WORLD COCOA AREA (HA)TRENDS FROM 1961 TO 2023

Table 4.5 Average World Cocoa Area (Ha) Trends from 1961 to 2023

TIME INTERWE L	AVERAG E OF BRAZIL	AVERAGE OFCAMEROO N	AVERAGE OF CÔTE D'IVOIRE	AVERAGE OF ECUADOR	AVERAG E OF GHANA	AVERAGE OF INDONESI A	AVERAG E OF NIGERIA
1961-1970	462131.7	380900	332700	211461.7	1605500	8139.4	700000
1971-1980	445359.9	390200.273	591470	239593.1	1325000	11175.4	700000
1981-1990	616085.2	415469	1184910	298824.3	836497.7	66511.4	702300
1991-2000	704509.3	358248.1	1726386.6	331650.5	1011989	401268.5	766150
2001-2010	631736.4	491974.9	2130987.8	367703.9	1621570	1125500.8	1169835
2011-2023	642612.4	653723.07	3728515.6	464806.4	1593937	1619720.6	1076198

SOURCE: FAOSTAT (FOOD AND AGRICULTURE ORGANIZATION OF THE UNITED NATIONS).

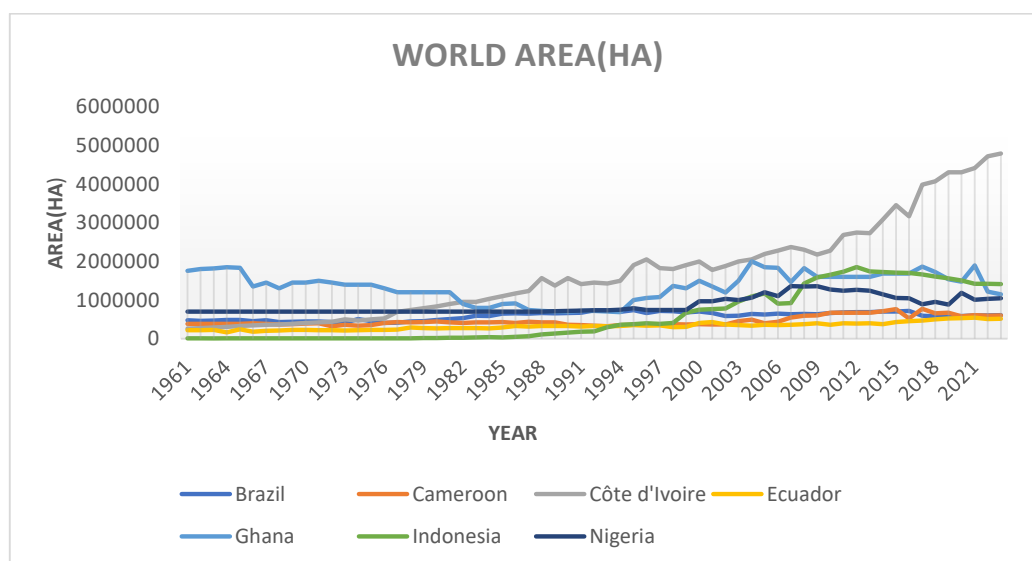


FIG 4.7 WORLD COCOA AREA TRENDS FROM 1961 TO 2023

The world cocoa area from 1961 to 2023 shows a clear transformation in global production patterns, reflecting economic, environmental, and policy-driven changes. In the early decades (1960s–1970s), Brazil dominated global cocoa cultivation with extensive plantation areas; however, its production area steadily declined due to the witches’ broom disease, adverse climatic conditions, and a shift toward more profitable crops such as soybeans and sugarcane. Meanwhile, Côte d’Ivoire (Ivory Coast) began a steady expansion in the late 1970s, rapidly accelerating after the 1980s to become the world’s largest cocoa producer, surpassing 6 million hectares by 2023.

This remarkable growth was driven by favorable government policies, export incentives, and rising international demand for cocoa.

Ghana, another West African nation, maintained a relatively stable but upward trend throughout the period, supported by strong government control, rehabilitation of old farms, and efforts to maintain bean quality. Indonesia emerged as a major Asian producer from the 1980s onward, driven by replanting programs and smallholder expansion, though it faced recent declines due to pests, diseases, and conversion of land to palm oil. Ecuador, known for its fine-flavor cocoa, saw steady growth, especially after 2000, reflecting its growing share in the premium cocoa market. Nigeria and Cameroon exhibited modest increases in cocoa area, but challenges such as aging plantations, limited access to finance, and inadequate infrastructure restricted large-scale growth.

Overall, the data reveal a geographical shift in cocoa cultivation from Latin America to West Africa, where Côte d'Ivoire and Ghana now dominate global cocoa area and production. The global increase in cocoa cultivation since the 1980s corresponds with rising global chocolate consumption, supportive agricultural policies, and improved farming practices. However, regional differences in growth also highlight the influence of disease, climate change, and economic diversification on national cocoa trends. This long-term pattern underscores West Africa's central role in sustaining the global cocoa supply chain.

COMPOUND ANNUAL GROWTH RATE (CAGR)

Table 4.6 CAGR of Cocoa Area in Major Producing Countries (1961–2023)

TIME INTERWEL	Brazil	Cameroon	Côte d'Ivoire	Ecuador	Ghana	Indonesia	Nigeria
1961-1981	0.16	0.56	6.11	1.43	-2.12	4.28	0
1982-2002	0.68	-0.89	3.71	1.42	2.89	20	1.44
2003-2023	-0.27	1.6	4.88	2.51	-0.96	2	-1

SOURCE: FAOSTAT (FOOD AND AGRICULTURE ORGANIZATION OF THE UNITED NATIONS).

The CAGR of cocoa area across the three time intervals shows clear structural changes in land expansion and contraction among major producing countries.

During 1961–1981, Côte d'Ivoire recorded strong area expansion (6.11%), indicating rapid conversion of land into cocoa and its emergence as a leading producer. Cameroon

and Ecuador showed moderate increases in area, while Brazil and Indonesia expanded only marginally. In contrast, Ghana experienced a contraction in cocoa area (−2.12%), reflecting institutional constraints and low incentives, and Nigeria showed stagnation with no net area growth.

In 1982–2002, Indonesia witnessed an exceptional surge in cocoa area (20%), driven by aggressive smallholder expansion and government support, making it a major global producer. Côte d’Ivoire and Ghana continued to expand area at a moderate pace, while Brazil showed slight improvement. Cameroon recorded a decline in cocoa area (−0.89%), suggesting structural and policy-related challenges. Ecuador and Nigeria posted modest area growth.

During 2003–2023, the pattern shifted toward area stabilization and contraction. Côte d’Ivoire remained resilient with continued area expansion (4.88%), while Ecuador and Cameroon showed moderate increases. However, cocoa area declined in Brazil, Ghana, and Nigeria, reflecting land saturation, aging plantations, and competition from alternative crops. Indonesia’s area growth slowed sharply to 2%, indicating maturation of its cocoa sector.

Overall, the area-based CAGR analysis indicates a transition from extensive land expansion in the early and middle periods to stabilization or decline in recent years, highlighting increasing land constraints and the need for productivity-driven growth in the global cocoa sector.

WORLD COCOA PRODUCTION(MT) TRENDS FROM 1961 TO 2023

Table 4.7 Average World Cocoa Production (MT) Trends from 1961 to 2023

TIME INTERWEL	AVERAG E OF BRAZIL	AVERAGE OFCAMEROO N	AVERAGE OF CÔTE D'IVOIRE	AVERAG E OF ECUADO R	AVERAG E OF GHANA	AVERAGE OF INDONESIA	AVERAG E OF NIGERIA
1961-1970	167942	92955.6	137148.1	51570.1	421930	1191.7	230750
1971-1980	258188	115441.1	276174.1	74629.4	344140	4587.2	197770
1981-1990	363886	119760.1	605134.4	84232.7	222787	52705.4	184180
1991-2000	283498	115896.8	1042983.3	80928.1	350619	308582.2	296600
2001-2010	200491	181477.1	1306792.8	92833.1	607611	725908.2	391530
2011-2023	263653	293065.437	1927771.5	231274	835467	685271.64	310985

SOURCE: FAOSTAT (FOOD AND AGRICULTURE ORGANIZATION OF THE UNITED NATIONS).

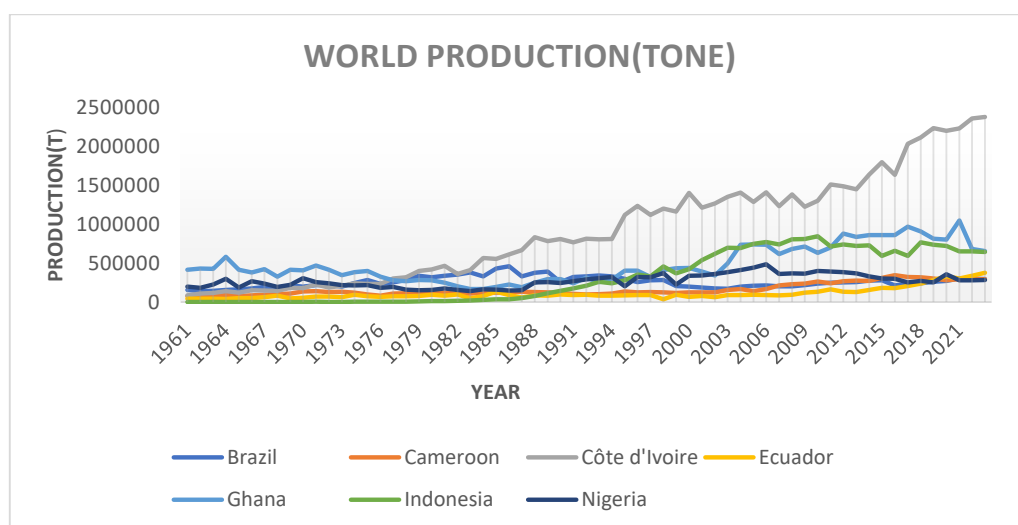


FIG 4.8 WORLD COCOA PRODUCTION TRENDS FROM 1961 TO 2023

The long-term trend in cocoa output among the top seven producing countries Brazil, Cameroon, Côte d'Ivoire, Ecuador, Ghana, Indonesia, and Nigeria from 1961 to 2023 shows a major transformation in global cocoa production patterns, with dominance shifting from Latin America to West Africa.

During the early years (1960s-1970s), Brazil and Ghana were the leading producers, contributing a major share of global cocoa output. Brazil's production remained strong until the 1980s but declined sharply afterward due to the witches' broom disease, adverse climatic conditions, and a shift toward more profitable crops such as soybeans and sugarcane. Ghana, though experiencing fluctuations, maintained relatively steady

output supported by effective government regulation, quality control through the Ghana Cocoa Board, and consistent farmer participation.

From the late 1970s onward, Côte d'Ivoire (Ivory Coast) began a rapid expansion in cocoa production, overtaking all other countries to become the world's largest producer. Its output rose dramatically, crossing 2.5 million tonnes by 2023. This exceptional growth was driven by favorable climate, government export incentives, and the active involvement of smallholder farmers, which positioned Côte d'Ivoire as the global hub of cocoa production.

Indonesia entered the global cocoa market strongly during the 1980s, showing a steep rise in production due to national replanting programs, technological improvements, and the adoption of high-yielding hybrid varieties. However, its production stabilized and slightly declined after 2010 as a result of pest infestations, aging trees, and competition from oil palm cultivation. Ecuador demonstrated continuous growth, especially after 2000, supported by increasing international demand for fine-flavor cocoa varieties such as *Arriba Nacional*.

Nigeria and Cameroon showed modest upward trends throughout the period, reflecting gradual modernization and improvements in farming practices, though their overall growth remained limited by low productivity, inadequate infrastructure, and restricted access to finance and inputs.

Overall, total world cocoa production exhibited a sharp upward trend, increasing several times over since 1961. This global rise has been fueled by rising chocolate consumption, favorable export-oriented agricultural policies, technological progress in crop management, and the shift of production to low-cost, high-yield regions in West Africa. At the same time, climate change, disease outbreaks, and economic diversification have contributed to regional variations in growth patterns.

In conclusion, from 1961 to 2023, global cocoa production evolved from being Latin America-centered to West Africa-dominated, with Côte d'Ivoire and Ghana emerging as the key producers. This steady increase reflects the combined effects of global market demand, improved agricultural technologies, policy support, and the resilience of smallholder farmers in maintaining the global cocoa supply chain.

COMPOUND ANNUAL GROWTH RATE (CAGR)

Table 4.8 CAGR of Cocoa Production (1961–2023)

TIME INTERWEL	Brazil	Cameroon	Côte d'Ivoire	Ecuador	Ghana	Indonesia	Nigeria
1961-1981	4.4	2.01	2.01	3.55	-2.78	14	-1.6
1982-2002	-3.61	0.32	5.89	-0.92	4.68	19.86	4.64
2003-2023	2.19	3.62	3.52	8.12	1.44	-0.72	-2.44

SOURCE: FAOSTAT (FOOD AND AGRICULTURE ORGANIZATION OF THE UNITED NATIONS).

The CAGR analysis reveals clear inter-period shifts in cocoa growth among major producing countries, shaped by structural conditions, policy responses, and market forces.

During 1961–1981, cocoa production largely expanded across countries, with Brazil, Cameroon, Côte d'Ivoire, and Ecuador recording moderate positive growth, indicating favorable prices and area expansion. In contrast, Ghana and Nigeria experienced negative growth due to institutional weaknesses and aging plantations, while Indonesia showed exceptionally rapid expansion, reflecting its late entry and aggressive smallholder-led growth.

In 1982–2002, growth became more uneven. Brazil's sharp decline reflects disease outbreaks and structural collapse, while Côte d'Ivoire and Ghana strengthened their global dominance through policy reforms and expansion. Indonesia maintained very high growth, and Nigeria recovered moderately, whereas Cameroon stagnated and Ecuador faced a mild slowdown.

During 2003–2023, growth moderated and signs of saturation emerged. Ecuador became the strongest performer due to quality-focused, export-oriented strategies. Cameroon, Côte d'Ivoire, and Brazil showed stable but moderate growth, while Ghana's expansion slowed due to land and productivity constraints. Indonesia and Nigeria entered decline, indicating structural fatigue from aging trees and rising production costs.

Overall, the results indicate a transition from early expansion to restructuring and recent stabilization or decline, with Indonesia's rise and slowdown and Ecuador's recent resurgence marking key shifts in the global cocoa economy.

WORLD COCOA YIELD(KG/HA) TRENDS FROM 1961 TO 2023

Table 4.9 Average World Cocoa Yield (Kg/Ha) Trends from 1961 to 2023

TIME INTERWEL	AVERAGE OF BRAZIL	AVERAGE OFCAMEROON	AVERAGE OF CÔTE D'IVOIRE	AVERAGE OF ECUADOR	AVERAG E OF GHANA	AVERAGE OF INDONESIA	AVERAGE OF NIGERIA
1961-1970	364.74	244.04	408.75	246.09	264.2	145.65	329.63
1971-1980	579.38	300.39	466.41	313.3	257.49	389.76	282.53
1981-1990	594.13	288.63	505.29	283.02	275.47	757.01	261.98
1991-2000	402.06	322.9	599.02	245.53	356.35	829.88	389.06
2001-2010	317.06	364.76	617.37	253.57	370.6	673.31	340.2
2011-2023	414.79	454.53	520.02	485.56	525.70	425.48	287.88

SOURCE: FAOSTAT (FOOD AND AGRICULTURE ORGANIZATION OF THE UNITED NATIONS).

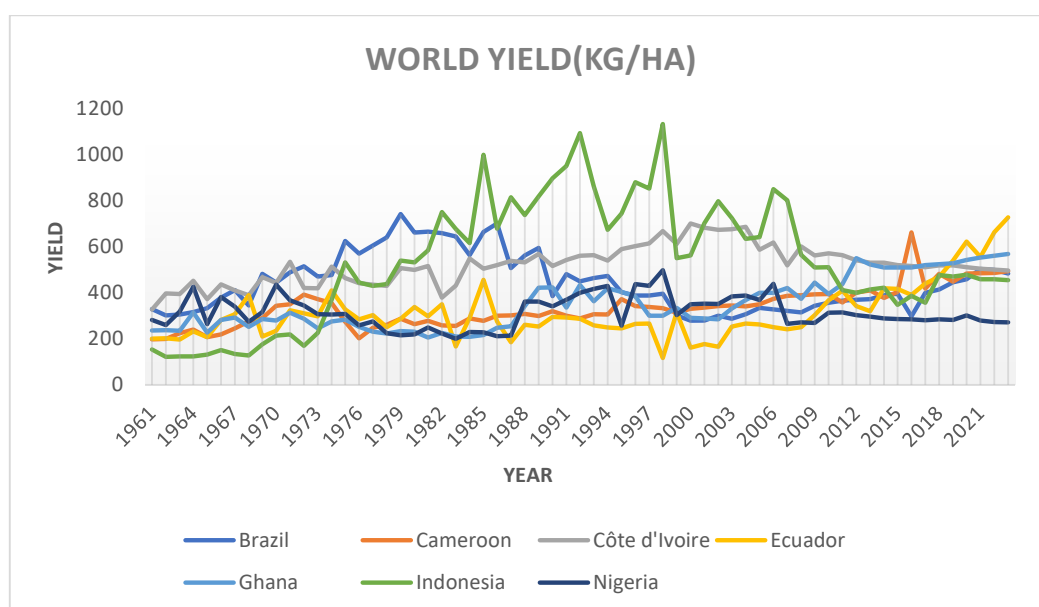


FIG 4.9 WORLD COCOA YIELD TRENDS FROM 1961 TO 2023

The long-term trend in cocoa yield (kg/ha) from 1961 to 2023 among the major producing countries Brazil, Cameroon, Côte d'Ivoire, Ecuador, Ghana, Indonesia, and Nigeria reveals significant fluctuations and gradual improvements over time, reflecting the combined effects of technological progress, climate variability, and policy interventions. Cocoa yield, which measures production efficiency per hectare, serves as a key indicator of agricultural productivity and sustainability in each producing nation.

In the early decades (1960s-1970s), yields were relatively low across all countries, generally between 200 and 400 kg/ha, due to the prevalence of traditional farming systems, minimal fertilizer use, and weak pest and disease management. During this period, Côte d'Ivoire and Ghana maintained moderate yields through structured marketing systems and government oversight, though productivity remained limited. From the late 1970s through the 1990s, Brazil and Indonesia recorded substantial improvements in yield, often surpassing 700 kg/ha, driven by modern cultivation techniques, hybrid varieties, and government-led replanting programs. Indonesia, in particular, achieved peaks above 1,000 kg/ha during the 1980s–1990s, making it one of the most productive regions for cocoa at that time. However, after 2010, its yields began to decline due to aging plantations, pest infestations such as the cocoa pod borer, and the conversion of farmland to oil palm cultivation.

Côte d'Ivoire, while leading the world in cocoa production volume, showed only moderate yield levels (400-600 kg/ha) because its production growth came largely from expanding cultivated area rather than from yield enhancement. Ghana's yield remained fairly stable but began to improve gradually in the 2010s as a result of government rehabilitation programs, fertilizer subsidies, and improved farm management practices.

Ecuador showed a notable upward trend after 2010, with yields rising sharply to around 700 kg/ha by 2023. This improvement is linked to increased investment in fine-flavor cocoa varieties (Arriba Nacional), better training for farmers, and sustainable farming initiatives supported by both the government and private sector. In contrast, Nigeria and Cameroon continued to exhibit relatively lower yields, constrained by limited access to inputs, old tree stock, and insufficient infrastructure, though both countries have shown small but steady progress in recent years.

Overall, the global cocoa yield trend from 1961 to 2023 reflects a gradual upward movement with significant differences across regions. Countries that invested in research, improved planting materials, and extension services such as Indonesia, Ghana, and Ecuador achieved higher productivity while others relying on area expansion experienced slower yield growth. The pattern also highlights the vulnerability of cocoa production to climate change, pest outbreaks, and market constraints.

In conclusion, the global cocoa yield pattern over the past six decades demonstrates a transition from low-input, traditional systems to more technologically advanced and

managed cultivation. While West Africa remains the global hub of production, sustained improvements in yield depend on continued investment in research replanting programs, climate-resilient varieties, and farmer training. These factors together will determine the long-term sustainability and competitiveness of the world cocoa sector.

COMPOUND ANNUAL GROWTH RATE (CAGR)

Table 4.10 CAGR of Cocoa Yield (1961–2023)

TIME INTERWEL	Brazil	Cameroon	Côte d'Ivoire	Ecuador	Ghana	Indonesia	Nigeria
1961-1981	4.58	1.43	1.35	2.08	-0.6	9.33	-1.6
1982-2002	-4.26	1.22	2.1	-2.32	1.74	-0.12	3.15
2003-2023	2.47	1.98	-1.3	5.47	2.44	-2.66	-1.45

SOURCE: FAOSTAT (FOOD AND AGRICULTURE ORGANIZATION OF THE UNITED NATIONS).

The CAGR of cocoa yield across the three time intervals highlights significant technological and structural differences among major cocoa-producing countries.

During 1961–1981, yield growth was largely positive in Brazil (4.58%), Ecuador (2.08%), Cameroon (1.43%), and Côte d'Ivoire (1.35%), indicating improvements in cultivation practices and favorable growing conditions. Indonesia recorded very strong yield growth (9.33%), reflecting rapid adoption of improved varieties and better farm management. In contrast, Ghana (-0.6%) and Nigeria (-1.6%) experienced declining yields, suggesting problems related to aging trees, limited input use, and weak extension support.

In the 1982–2002 period, yield performance became more uneven. Brazil recorded a sharp decline (-4.26%), mainly due to disease outbreaks and reduced farm investment. Ecuador also experienced yield contraction (-2.32%), reflecting transitional challenges. Meanwhile, Côte d'Ivoire (2.1%), Ghana (1.74%), Cameroon (1.22%), and Nigeria (3.15%) showed yield improvements, indicating better management practices and partial technological gains. Indonesia's yield remained almost stagnant (-0.12%), suggesting the early signs of yield saturation after rapid initial growth.

During 2003–2023, yield trends again shifted. Ecuador emerged as the strongest performer (5.47%), driven by improved varieties and quality-focused production systems. Brazil (2.47%), Ghana (2.44%), and Cameroon (1.98%) recorded moderate yield growth, indicating recovery and productivity enhancement. In contrast, Côte d'Ivoire (-1.3%), Indonesia (-2.66%), and Nigeria (-1.45%) experienced declining

yields, reflecting aging plantations, pest and disease pressure, and rising production costs.

Overall, the yield-based CAGR analysis shows an initial phase of technological gains, followed by mixed performance and recent productivity challenges, emphasizing the growing importance of replanting, technology adoption, and yield-led growth for sustaining cocoa production.

WORLD COCOA AREA ,PRODUCTIONAND YIELD TRENDS FROM 1961 TO 2023

Table 4.11 Average World Cocoa Area, Production, and Yield (1961–2023)

TIME INTERWEL	AREA(HA)	PRODUCTION(MT)	YIELD(KG/HA)
1961-1970	4351855.5	1339783.1	308.41
1971-1980	4449922.6	1531459.9	344.02
1981-1990	5141322.4	2069045.7	399.46
1991-2000	6361213.2	2939977.909	462.59
2001-2010	8482711.1	3934433.847	464.48
2011-2023	11096467.69	5140083.532	462.3846

SOURCE: FAOSTAT (FOOD AND AGRICULTURE ORGANIZATION OF THE UNITED NATIONS).

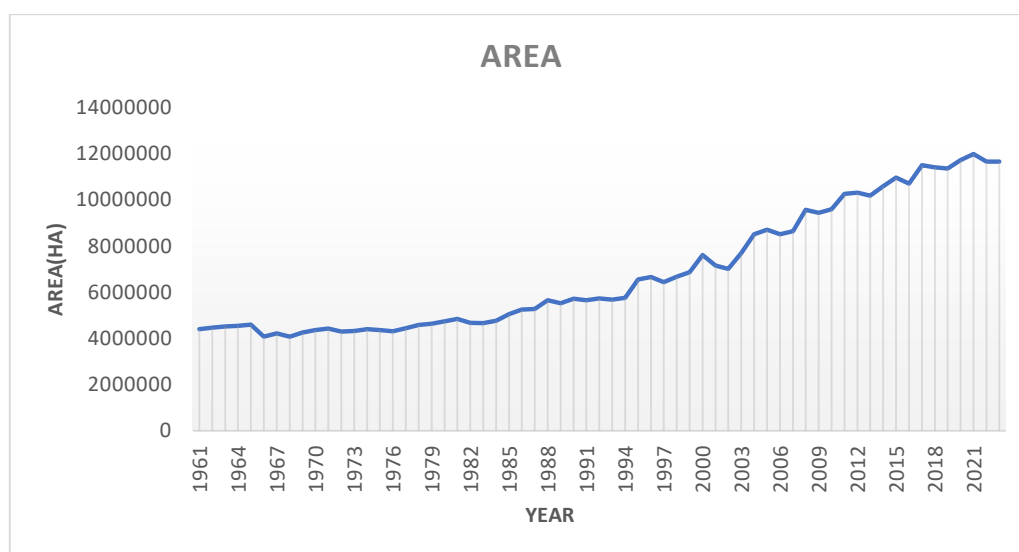


FIG 4.10 WORLD COCOA AREA TRENDS FROM 1961 TO 2023

The graph illustrates the global cocoa cultivation area from 1961 to 2023, showing a consistent long-term expansion in land devoted to cocoa production. In the early 1960s and 1970s, the area remained relatively stable around 4 to 5 million hectares, reflecting limited investment and traditional farming methods in major cocoa-growing regions. From the late 1980s onward, there was a steady upward trend as countries in West

Africa particularly Côte d'Ivoire and Ghana expanded cocoa plantations to meet rising international demand and take advantage of favorable export prices. The most significant growth occurred between 1990 and 2015, when global consumption of chocolate and cocoa-based products surged, prompting many tropical countries to convert additional agricultural land for cocoa cultivation. After 2015, the growth rate slowed slightly, with the area stabilizing around 12 million hectares, likely due to land-use constraints, sustainability concerns, and replanting efforts replacing expansion. Overall, the increase in cocoa area reflects both economic incentives from the global chocolate industry and the agricultural shift toward high-value export crops in developing nations.

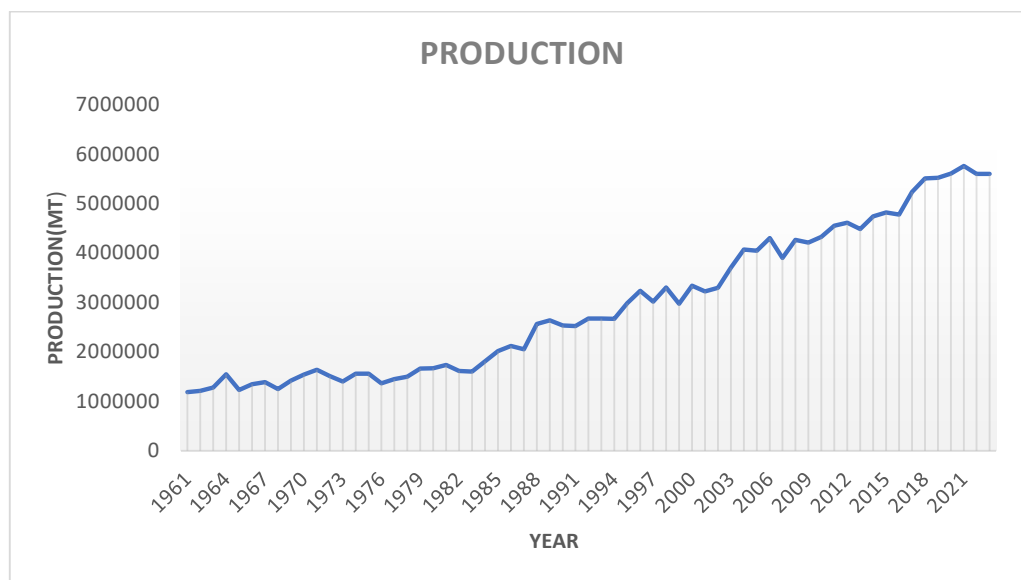


FIG 4.11 WORLD COCOA PRODUCTION TRENDS FROM 1961 TO 2023

The graph shows the trend of global cocoa production from 1961 to 2023, highlighting a significant and steady increase over the decades. In the early 1960s, cocoa production was around 1.2 million tonnes, reflecting the limited scale of cultivation and traditional farming techniques. From the late 1970s onwards, production began to rise gradually as new cocoa plantations were established, particularly in West African nations such as Côte d'Ivoire, Ghana, and Nigeria, which benefited from favorable climatic conditions and growing export markets. A more pronounced surge occurred after 1990, driven by expanding global demand for chocolate, improvements in agricultural practices, and the introduction of high-yielding cocoa varieties. The period between 2000 and 2018 saw the fastest growth, with global production surpassing 5 million

tonnes, fueled by both area expansion and yield improvements. However, in recent years, production appears to have plateaued slightly, possibly due to aging trees, climate variability, and sustainability concerns limiting further expansion. Overall, the long-term upward trend reflects technological advancement, increased investment, and the rising global appetite for cocoa-based products.

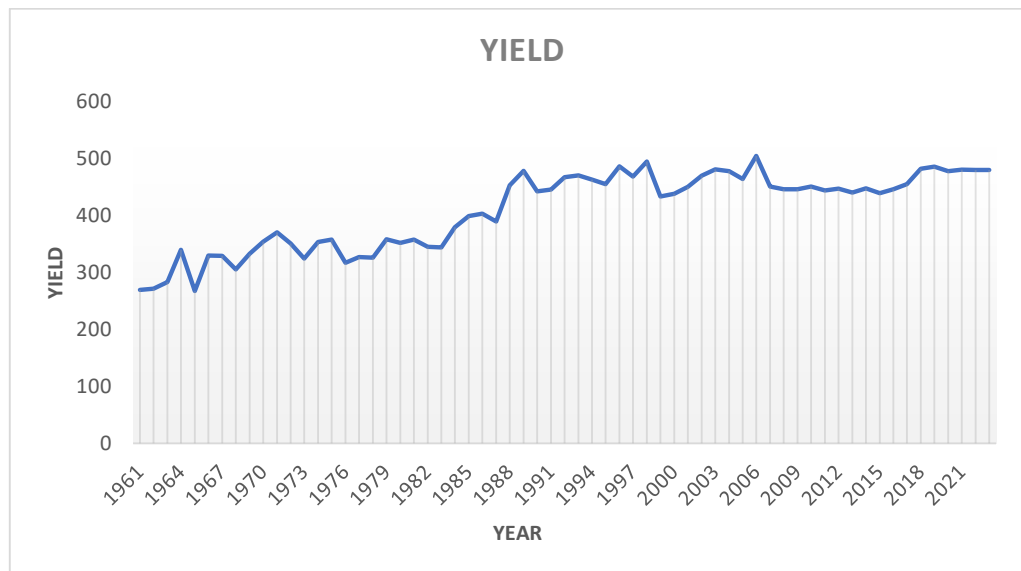


FIG 4.12 WORLD COCOA YIELD TRENDS FROM 1961 TO 2023

The graph depicts the global cocoa yield trend from 1961 to 2023, showing moderate improvement with noticeable fluctuations over time. In the early 1960s, the yield was around 270 kg per hectare, reflecting traditional cultivation methods, limited fertilizer use, and low adoption of improved crop varieties. Between the 1970s and late 1980s, yields gradually increased as better farming practices, disease-resistant cocoa plants, and improved agricultural inputs were introduced in major producing countries. The late 1980s to early 2000s marked a period of relatively higher yields, peaking around 500 kg per hectare, largely due to intensified farming and government-supported cocoa development programs in West Africa and Latin America. However, after 2005, yields exhibited a plateau with minor fluctuations, suggesting challenges such as aging cocoa trees, soil degradation, pest infestations, and inconsistent rainfall patterns linked to climate change. Despite the overall increase compared to the 1960s, the yield has remained relatively stagnant in recent decades, indicating that future gains in cocoa production may rely more on technological innovation, replanting programs, and sustainable agricultural practices rather than area expansion alone.

COMPOUND ANNUAL GROWTH RATE (CAGR)

Table 4.12 CAGR of Area, Production, and Yield (1961–2023)

Time interval	AREA(HA)	PRODUCTION(MT)	YIELD(KG/HA)
1961-1981	0.28%	1.39%	1.11%
1982-2002	2.28%	3.58%	1.26%
2003-2023	2.04%	2.14%	0.10%
TOTAL	1.94	2.8	0.83

SOURCE: FAOSTAT (FOOD AND AGRICULTURE ORGANIZATION OF THE UNITED NATIONS).

The CAGR analysis of world cocoa area, production, and yield reveals important structural changes over time. During 1961–1981, global cocoa production growth (1.39%) was driven more by yield improvement (1.11%) than by area expansion (0.28%), indicating early technological gains and better farm management practices. In the 1982–2002 period, cocoa experienced its most dynamic phase, with strong expansion in area (2.28%) and production (3.58%), while yield growth (1.26%) remained moderate, suggesting that production increases were achieved through a combination of land expansion and productivity improvements. In contrast, during 2003–2023, although cocoa area continued to expand (2.04%), production growth slowed to 2.14% and yield growth nearly stagnated (0.10%), reflecting saturation effects, aging plantations, and rising biotic and economic constraints. Overall, the total period results show that global cocoa production growth (2.8%) has been driven primarily by area expansion (1.94%), with relatively weaker contributions from yield improvement (0.83%), highlighting the increasing need for productivity-led growth through replanting, improved varieties, and better agronomic practices in the world cocoa sector.

EXPORT AND IMPORT PERFORMANCE OF COCOA

Table 4.13 Average Export And Import Performance Of Cocoa

TIME INTERWEL	AVERAGE EXPORT		AVERAGE IMPORT	
	Quantity(MT)	Value(Rs. in Crores)	Quantity(MT)	Value(Rs. in Crores)
1997-2006	1586	15.52	6209	57.8
2007-2016	13567	355	39074	753.3
2017-2025	28094.4	1376.6	98175	3070.5

SOURCE : DIRECTORATE OF CASHEWNUT AND COCOA DEVELOPMENT

EXPORT OF COCOA OVER THE YEAR



FIG 4.13 EXPORT QUANTITY OF COCOA OVER THE YEAR

The graph illustrates the trend of India's cocoa export quantity from 1996–1997 to 2024–2025, showing a remarkable growth pattern with fluctuations in recent years. During the early period (1996–2005), cocoa exports were minimal due to the limited production base and weak international demand for Indian cocoa. However, from 2006 onwards, exports began to rise steadily as domestic processing industries expanded and quality standards improved.

A significant surge occurred between 2010–2015, when export quantities increased sharply, peaking around 2015–2016 at over 30,000 MT. This growth was driven by strong global demand, India's improved processing capacity, and increased participation in international trade networks. The export boom also reflected the success of Indian companies in exporting cocoa butter, powder, and chocolate products to regions such as the Middle East and Southeast Asia.

After 2016, exports showed periodic fluctuations, influenced by global price volatility, domestic consumption growth, and supply limitations. However, another notable increase was seen between 2021–2024, reflecting recovery after global trade disruptions and rising international demand for processed cocoa.

Overall, the trend highlights India’s emergence as a growing exporter of cocoa products, supported by improved processing technology and global market integration. Yet, maintaining consistent export growth will depend on enhancing domestic raw cocoa availability and sustainable farming practices.



FIG 4.14 EXPORT VALUE OF COCOA OVER THE YEAR

The graph represents the trend of India’s cocoa export value from 1996–1997 to 2024–2025, indicating a substantial increase over the years, especially after 2010. During the early period (1996–2005), export values were negligible due to limited production, lack of export infrastructure, and low international competitiveness. A steady upward trend began after 2006–2007, coinciding with the expansion of India’s cocoa processing sector and improved quality standards in cocoa-based products.

From 2010–2015, export value grew sharply as global demand for cocoa butter, powder, and chocolate products surged. The highest rise occurred between 2013–2016, driven by favorable international prices and the diversification of Indian cocoa exports into value-added forms rather than raw beans. Despite moderate fluctuations from 2017–2022, export value remained consistently high, supported by growing demand from Asian and Middle Eastern markets.

A remarkable spike is observed in 2024–2025, with export value surpassing ₹2500 crore, indicating record growth. This sharp rise can be attributed to increased global cocoa prices, post-pandemic demand recovery, and India’s greater participation in international cocoa trade. Overall, the trend reflects India’s transition from a small-scale exporter to a significant player in the global cocoa market, emphasizing the importance of quality enhancement and sustainable production for future growth.

IMPORT OF COCOA OVER THE YEAR



FIG 4.15 IMPORT QUANTITY OF COCOA OVER THE YEAR

The graph shows the trend of India’s cocoa import quantity from 1996–1997 to 2024–2025, indicating a strong upward growth over the years. During the early period (1996–2005), imports remained very low due to limited domestic demand and small-scale chocolate production. From 2006 onwards, imports started to rise steadily as the confectionery, bakery, and beverage industries expanded rapidly in India.

A sharp increase is observed after 2010–2011, when imports crossed 40,000 MT, mainly due to rising consumer demand, industrial use of cocoa, and insufficient domestic production to meet needs. Between 2015 and 2020, imports grew further with the entry of global chocolate brands, increased processed food exports, and higher use of cocoa butter and powder in cosmetics and pharmaceuticals.

From 2020–2021 to 2023–2024, imports peaked at around 120,000 MT, reflecting strong dependency on international suppliers like Ghana, Côte d’Ivoire, and Indonesia.

The slight dip in 2024–2025 may suggest stable demand or increased domestic cultivation efforts.

Overall, the graph highlights India’s rising cocoa import dependency, driven by rapid industrial growth and limited local supply, emphasizing the need for enhanced domestic production and sustainable cocoa farming practices to balance future demand.

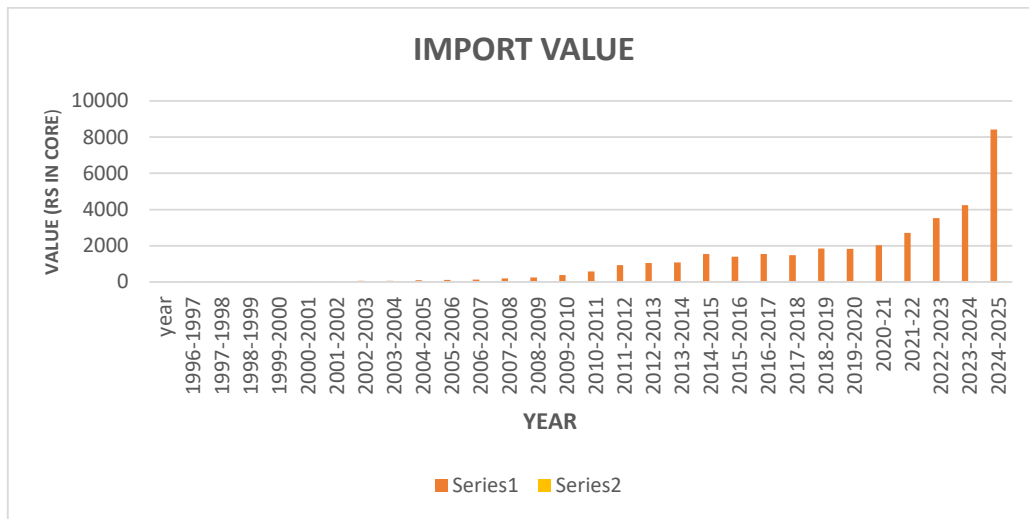


FIG 4.16 IMPORT VALUEOF COCOA OVER THE YEAR

The graph depicts the trend of India’s cocoa import value from 1996–1997 to 2024–2025, showing a sharp and consistent rise, especially in the last decade. In the initial years (1996–2005), import values were minimal due to low domestic consumption and a limited chocolate manufacturing industry. However, from 2006 onwards, the import value began to grow steadily, driven by the expansion of confectionery industries, increasing demand for processed cocoa products, and rising consumer preference for chocolate-based items.

A significant surge occurred after 2010–2011, as India’s food processing sector modernized, and multinational brands entered the market, increasing the need for high-quality imported cocoa beans, butter, and powder. Between 2015 and 2021, the import value rose moderately, supported by urbanization, rising disposable incomes, and greater use of cocoa in beverages, bakery products, and cosmetics.

The graph shows a dramatic spike from 2021–2022 to 2024–2025, with the import value peaking around ₹8,500 crore, reflecting a combination of higher global cocoa

prices, currency fluctuations, and growing domestic demand. This surge also indicates limited local production capacity, forcing India to rely heavily on imports from major producers such as Côte d'Ivoire, Ghana, and Indonesia. Overall, the trend highlights India's increasing dependency on imported cocoa, emphasizing the need for enhanced domestic cultivation and value-added processing to reduce import costs.

PRICE ANALYSIS

FROM YEARLY MARKET PRICE OF COCOA IN KERALA

Table 4.14 Average Monthly Price Behavior of Cocoa in Kerala

TIME INTERWEL	AVERAGE PRICE
1999-2008	1975
2009-2018	3926
2019-2025	26202

SOURCE: DIRECTORATE OF ECONOMICS AND STATISTICS (DES), GOVERNMENT OF KERALA.



FIG 4.17 YEARLY WHOLE SALE PRICE OF COCOA IN KERALA

The trend in Kerala's yearly cocoa prices (₹/quintal) shows a prolonged period of relative stability followed by a sharp and unprecedented escalation in recent years. From the late 1990s to around 2010, cocoa prices remained low and largely stable at below ₹5,000 per quintal, reflecting limited domestic demand, weak integration with international cocoa markets, and adequate supply conditions. During the 2011–2019 period, prices exhibited a gradual upward trend, mostly ranging between ₹3,000 and ₹5,000 per quintal, indicating improving market linkages and rising demand from domestic processors, though still moderated by supply-side constraints. A clear structural break emerges after 2020, when cocoa prices surged dramatically, reaching exceptionally high levels of ₹15,000-₹70,000 per quintal by 2023-2025. This extraordinary rise mirrors the transmission of global cocoa price shocks into the Kerala market, driven by international supply shortages, climate-induced production losses in

major producing countries, rising global demand, currency movements, and speculative pressures. Overall, the yearly price trend highlights Kerala's transition from a relatively stable, locally driven cocoa price regime to a highly volatile, globally influenced market, emphasizing the need for effective price-risk management and policy interventions to safeguard farmers while enabling them to capitalize on favorable price movements.

SEASONAL INDICES

Table 4.15 Seasonal Index and Monthly Price Behavior of Cocoa in Kerala

MONTH	MONTHLY AVERAGE	SEASONAL INDEX	INTERPRETATION
January	8312	112.73	ABOVE AVERAGE
February	8788	119.19	ABOVE AVERAGE
March	8578	116.34	ABOVE AVERAGE
April	9526	129.2	ABOVE AVERAGE
May	8678	117.69	ABOVE AVERAGE
June	6640	90.05	BELOW AVERAGE
July	5529	74.98	LOWEST MONTH
August	5471	74.2	LOWEST MONTH
September	5587	75.77	LOWEST MONTH
October	6172	83.71	LOWEST MONTH
November	7272	98.63	BELOW AVERAGE
December	7930	107.55	ABOVE AVERAGE

SOURCE: DIRECTORATE OF ECONOMICS AND STATISTICS (DES), GOVERNMENT OF KERALA.

The seasonal index analysis of Kerala cocoa prices reveals a pronounced and consistent intra-year price pattern driven by supply availability and market arrivals. Prices remain well above the annual average from January to May, with seasonal indices ranging from 112.73 in January to a peak of 129.20 in April, indicating strong price realization during the lean-arrival period when market supplies are relatively limited and demand from processors is steady. This high-price phase reflects post-harvest exhaustion of stocks from the previous season and increased procurement by traders. From June onward, prices begin to weaken, falling below the average in June (90.05) and reaching the lowest levels during July to October, with indices between 74.20 and 83.71, coinciding with the main harvest season and monsoon-induced market arrivals, which increase supply and exert downward pressure on prices. The recovery phase starts in November (98.63) and strengthens in December (107.55) as arrivals taper off and demand revives, leading to a price rebound. Overall, the seasonal

indices clearly demonstrate that Kerala cocoa prices are highly seasonal, with farmers receiving better prices in the pre-monsoon and early summer months, underscoring the importance of storage, timing of sales, and market intelligence to enhance farm income.

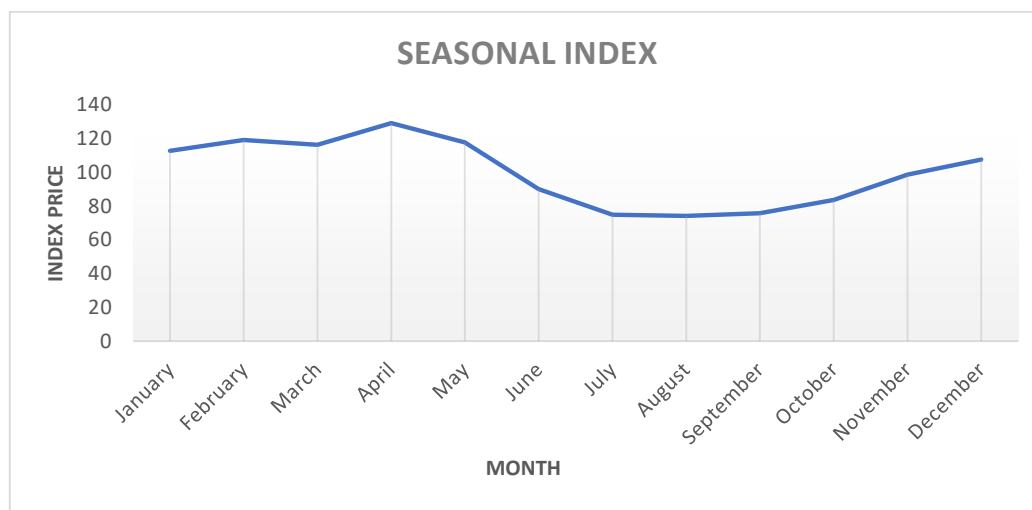


FIG 4.18 SEASONAL INDEX PRICE OF COCOA IN KERALA

The seasonal index pattern of Kerala cocoa prices exhibits a clear and well-defined annual price cycle, as illustrated by the chart. Prices remain consistently above the annual average during January to May, with the index rising from around 113 in January to a peak of about 129 in April, reflecting strong market prices during the lean supply period when arrivals are limited and demand from processors and traders is relatively high. From June onward, the index declines sharply, falling below 100 and reaching the lowest levels during July and August (around 74–75), which coincides with the peak harvest season and monsoon months, when increased market arrivals and weather-related selling pressure depress prices. The index remains low through September, after which a gradual recovery begins in October, indicating a reduction in arrivals and improving demand conditions. By November and December, the index climbs back above the annual average, signaling a price rebound toward the year-end. Overall, the seasonal index clearly demonstrates that cocoa prices in Kerala are highly seasonal, with significant implications for farmers' income; strategic storage and delayed sales during low-price months could help producers realize better price margins during the high-index periods.

DOMESTIC AND INTERNATIONAL PRICES

Table 4.16 Average Domestic and International Cocoa Prices

TIME INTERWEL	AVERAGE DOMESTIC PRICE(RS/KG)	AVERAGE INTERNATIONAL PRICE(RS/KG)
1994-2003	56.97	54.59
2004-2013	106.84	111.95
2014-2024	243.45	232.45

SOURCE : DIRECTORATE OF CASHEWNUT AND COCOA DEVELOPMENT

DOMESTIC PRICE OF COCOA

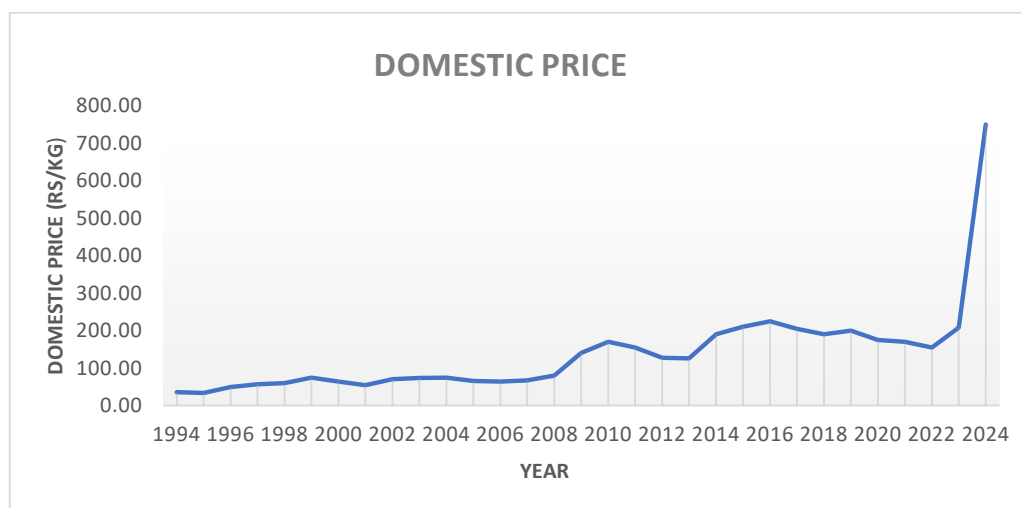


FIG 4.19 DOMESTIC PRICE OF COCOA

The domestic cocoa price trend from 1994 to 2024 shows a gradual increase followed by a sharp and unprecedented rise in recent years. During the period from 1994 to 2007, prices remained low and stable due to limited domestic demand and low production levels. From 2008 to 2012, prices began to increase moderately as the Indian chocolate and confectionery industries expanded, leading to higher demand for cocoa beans. Between 2013 and 2020, prices fluctuated but remained relatively steady as domestic production improved and imports helped balance supply. However, a significant surge occurred from 2022 to 2024, with prices reaching an all-time high of around ₹750 per kg. This sharp increase can be attributed to several key factors, including global cocoa shortages caused by unfavorable weather in major producing countries like Ghana and Ivory Coast, rising export demand, and higher domestic production costs. Additionally, pest infestations, climate-related yield reductions, and post-pandemic inflation have also contributed to the rapid price escalation. The overall

trend indicates that domestic cocoa prices in India have become highly influenced by global market fluctuations and supply chain disruptions, highlighting the need for sustainable production practices and improved resilience against global price volatility.

INTERNATIONAL PRICE OF COCOA



FIG 4.20 INTERNATIONAL PRICE OF COCOA

The international cocoa price trend from 1994 to 2024 shows a long-term upward movement, with a sharp spike in recent years. Between 1994 and 2006, cocoa prices remained relatively low and stable due to balanced supply from major producers like Ghana and Ivory Coast and moderate global demand. From 2007 to 2016, prices rose steadily, driven by growing demand from the global chocolate and confectionery industries, especially in emerging markets such as China and India. However, fluctuations during 2017-2021 reflected variable global supply conditions, currency movements, and changes in trade policies. The most dramatic price increase occurred from 2022 to 2024, where international prices surged sharply to around 750 per kg. This unprecedented rise was primarily caused by severe climate issues in West Africa extreme rainfall and drought cycles that reduced cocoa yields significantly. Additionally, increased pest infestations, aging cocoa trees, and reduced fertilizer use due to high costs further constrained production. On the demand side, post-pandemic recovery and rising global consumption of cocoa based products intensified the pressure on limited supplies. The combined impact of these factors led to a record-

high global cocoa price, reflecting the growing vulnerability of cocoa markets to environmental and economic disruptions.

COMPARISON OF DOMESTIC AND INTERNATIONAL PRICE OF COCOA

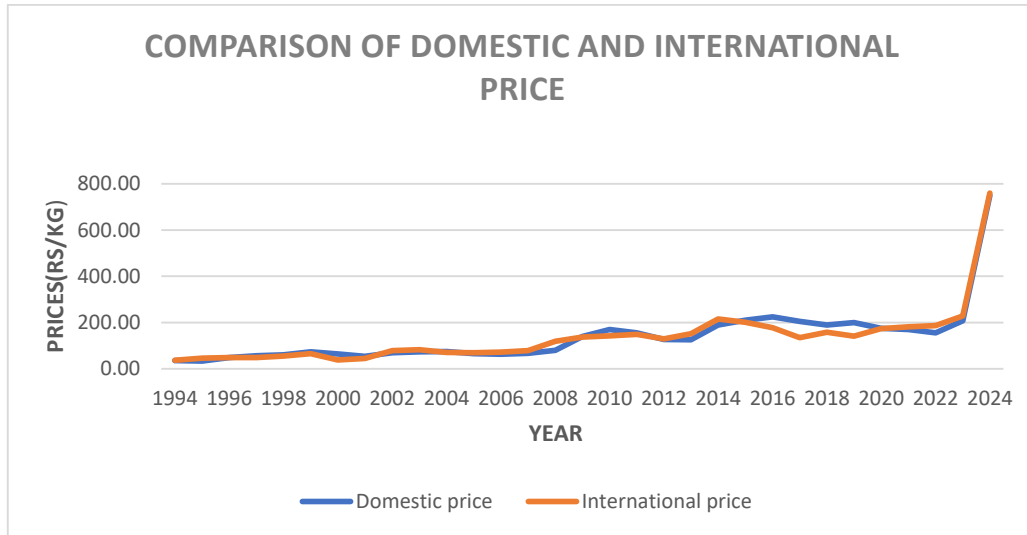


FIG 4.21 COMPARISON OF DOMESTIC AND INTERNATIONAL PRICE OF COCOA

The comparison graph of domestic and international cocoa prices (1994-2024) reveals a strong positive correlation between the two, indicating that global market dynamics directly influence India's cocoa pricing trends. From 1994 to around 2006, both domestic and international prices remained relatively low and stable, reflecting balanced production and moderate global demand. Between 2007 and 2016, prices began to rise gradually due to growing demand from the confectionery and cosmetics industries, alongside supply challenges in major cocoa-producing countries such as Ghana and Ivory Coast.

During 2017-2021, fluctuations in both price lines show the impact of market uncertainties, currency fluctuations, and variations in cocoa yield. Despite minor domestic stabilization due to internal supply adjustments, international price volatility continued to influence local markets. The most striking feature of the graph is the sharp and simultaneous surge from 2022 to 2024, where both domestic and global prices spiked dramatically international prices reaching nearly ₹750/kg equivalent.

This steep rise is mainly attributed to climate-induced production declines in West Africa, including droughts and crop diseases, combined with higher global consumption and increased transportation and input costs. As India relies on imports

to meet demand, domestic prices followed international trends, showing how tightly linked the two markets have become under global supply pressures.

PRICE INSTABILITY IN DOMESTIC AND INTERNATIONAL PRICE

Table 4.17 Price Instability Measures of Domestic and International Cocoa Prices

	INTERNATIONAL PRICE	DOMESTIC PRICE
MEAN	136.2	139.23
SD	129.37	129.22
CV	94.98%	92.81%
R SQR	0.44	0.46
CDVI	71.04	68.08
CORREALATION	0.98	

SOURCE : DIRECTORATE OF CASHEWNUT AND COCOA DEVELOPMENT

The analysis of international and domestic price movements reveals significant insights into their stability and interdependence. The mean price level is slightly higher for the domestic market (139.23) compared to the international market (136.2), indicating that domestic prices tend to remain marginally above global averages. The standard deviation (SD) values are almost identical for both series (129.37 and 129.22), suggesting comparable levels of dispersion in price behavior. However, the coefficient of variation (CV) 94.98% for international prices and 92.81% for domestic prices shows a high degree of overall variability in both series before adjusting for long-term trends.

When adjusted using the Cuddy Della Valle Instability Index (CDVI), the instability reduces to 71.04% for international prices and 68.08% for domestic prices, indicating a considerable but somewhat lower level of trend adjusted instability. This means that while both markets experience fluctuations, domestic prices are relatively more stable compared to international prices. The lower domestic instability can be attributed to government interventions such as price regulation, subsidies, and import controls, which cushion the domestic market against global shocks. Conversely, the international price series is more exposed to global demand–supply dynamics, speculative trading, and external shocks such as geopolitical tensions or climatic events affecting major producers.

The correlation coefficient of 0.98 between domestic and international prices demonstrates a very strong positive relationship, implying that price movements in the

international market are closely mirrored by the domestic market. This high degree of correlation suggests that domestic prices remain highly responsive to global price trends despite moderate internal stabilization mechanisms. The strong association likely arises from trade linkages, dependence on global commodity markets, and synchronized demand patterns. Overall, the results highlight that while domestic prices are slightly more stable, both price series are strongly interconnected, reflecting a high degree of market integration influenced by global economic forces.

COMPOUND ANNUAL GROWTH RATE INTERNATIONAL AND DOMESTIC PRICE

Table 4.18 Compound Annual Growth Rate of International and Domestic Cocoa Prices

TIME PERIOD	INTERNATIONAL	DOMESTIC
1994-2010	7.42%	6.81%
2011-2024	6.91%	5.80%

SOURCE : DIRECTORATE OF CASHEWNUT AND COCOA DEVELOPMENT

The comparison between the two time periods 1994-2010 and 2011-2024 shows distinct patterns in international and domestic price growth, each influenced by specific economic and policy factors. During 1994-2010, international prices grew by 7.42% and domestic prices by 6.81%, reflecting a period of strong globalization, trade liberalization, and market integration. Many economies, including developing ones, opened up to international trade, leading to higher demand for exports and increased price linkages between global and local markets. The 1990s and early 2000s also saw rising commodity demand driven by industrial growth in emerging economies like China and India, pushing international prices upward. Domestically, prices followed a similar trend due to high import dependency, weaker regulatory controls, and the influence of international price transmission mechanisms.

In contrast, the 2011-2024 period marks a slowdown in growth rates, with international prices rising by 6.91% and domestic prices by 5.80%. This moderation can be attributed to several global and national factors. Internationally, the aftermath of the 2008 global financial crisis, technological improvements in production, and supply chain stabilization reduced price volatility. On the domestic front, government interventions, such as price stabilization policies, subsidies, and tighter monetary controls, helped contain inflation. Additionally, exchange rate adjustments, growth in

domestic production, and import diversification reduced dependence on international markets. As a result, the gap between international and domestic growth widened from 0.61% to 1.11%, indicating weakened price transmission and a shift toward greater self-reliance and economic stability. Overall, the two periods represent a transition from global integration to controlled, sustainable growth.

COINTEGRATION

1. AUGMENTED DICKEY-FULLER (ADF) TEST RESULTS

Table 4.19 Augmented Dickey–Fuller (ADF) Test Results for Cocoa Price Series

Variable	ADF Stat	p-value	Used Lags	Interpretation
Year	-0.1213	0.9473	9	Non-stationary
International Price	3.7769	1	9	Non-stationary
Domestic Price	2.5906	0.9991	8	Non-stationary

SOURCE : DIRECTORATE OF CASHEWNUT AND COCOA DEVELOPMENT

The Augmented Dickey-Fuller (ADF) test examines whether a time series is stationary that is, whether its mean, variance, and autocovariance remain constant over time. The results indicate that all three variables Year, International Price, and Domestic Price are non-stationary. Each variable shows high p-values (all greater than 0.05), meaning the null hypothesis of non-stationarity cannot be rejected. This implies that the mean and variance of cocoa prices change with time, likely due to market forces, global supply-demand variations, policy interventions, and inflationary trends. Non-stationary data typically exhibit upward or downward trends rather than fluctuating around a constant mean. Therefore, before proceeding to any forecasting or regression modeling, it is necessary to either difference the data to achieve stationarity or test for cointegration, which checks if a stable long-term equilibrium exists between the non-stationary series. This forms the basis for long-run relationship analysis between domestic and global cocoa prices.

1. 2.ENGLE -GRANGER PAIRWISE COINTEGRATION TEST RESULTS

Table 4.20 Engle–Granger Cointegration Test Results for Cocoa Prices

Dependent (y)	Independent (x)	EG t-stat	EG p-value	Residual ADF	Residual p-value	Result
Year	International Price	-3.089	0.0905	-3.089	0.002	Cointegrate
Year	Domestic Price	-2.275	0.385	-2.275	0.022	Cointegrate
International Price	Domestic Price	-3.468	0.035	-3.468	0.0005	Cointegrate

SOURCE : DIRECTORATE OF CASHEWNUT AND COCOA DEVELOPMENT

The Engle–Granger test investigates whether two non-stationary variables share a long-run equilibrium relationship. The results reveal that all pairs of variables, particularly Domestic Price and International Price, are cointegrated at the 5% significance level. The test statistic for Domestic vs. International Price (-3.468) is significant with a p-value of 0.0005, which strongly rejects the null hypothesis of no cointegration. This means that while domestic and global cocoa prices may fluctuate in the short term, they move together over the long term, adjusting toward a shared equilibrium. In practical terms, international cocoa price changes tend to transmit into the domestic Indian cocoa market, indicating strong market integration and price transmission mechanisms. Additionally, the Year variable's cointegration with both price series suggests that time plays a structural role in the adjustment process, possibly reflecting global trade liberalization, agricultural reforms, or shifts in demand over the years. Overall, these results confirm that domestic prices are not isolated they respond to global market forces, demonstrating the interconnectedness of local and international cocoa economies.

3. JOHANSEN COINTEGRATION EIGENVECTORS

Table 4.21 Johansen Cointegration Test: Normalized Eigenvectors (CVs)

Variable	CV1	CV2	CV3
Year	0.3762	-0.0888	0.1059
International Price	-0.0606	-0.0463	0.0095
Domestic Price	0.0058	0.0447	-0.0372

SOURCE : DIRECTORATE OF CASHEWNUT AND COCOA DEVELOPMENT

The Johansen cointegration test provides a multivariate framework confirming whether more than two non-stationary series share one or more cointegrating relationships. The eigenvector results reveal that Year, International Price, and Domestic Price are all part of a long-run equilibrium system. The eigenvector coefficients show that the Year variable has the highest loading (0.3762 in CV1), suggesting that temporal trends strongly drive long-term price relationships. Meanwhile, the relatively smaller coefficients for International Price and Domestic Price indicate that their adjustments toward equilibrium occur gradually rather than abruptly. This aligns with economic reality domestic prices often lag behind international changes due to policy buffers, transaction delays, and market imperfections. The Johansen test therefore reinforces the Engle Granger findings by confirming a stable long-term relationship among all variables. This means that despite temporary shocks or short-run deviations, domestic and global cocoa prices consistently realign over time toward equilibrium, reflecting the robustness of global market linkages and trade efficiency.

CONCLUSION

The statistical analysis of world cocoa production and price trends reveals significant transformations in both global and domestic cocoa sectors over the past six decades. The study clearly demonstrates that cocoa cultivation has progressively shifted from Latin America to West Africa, where Côte d'Ivoire and Ghana have become dominant players, together accounting for more than 60% of global output. This regional concentration underscores the crucial role of African economies in sustaining global chocolate industries while highlighting vulnerabilities arising from dependency on a few major producers.

At the global level, the compound annual growth rate (CAGR) analysis shows consistent expansion in cocoa area and production, particularly during the 1980s and 1990s, though recent decades indicate slowing growth and emerging productivity challenges. The volatility in cocoa prices continues to be influenced by climatic conditions, supply shocks, and speculative market behavior, leading to cyclical fluctuations. Gilbert's and Ruf's findings confirm that both structural and speculative factors shape international price dynamics, and stabilization policies remain vital for maintaining farmer income security.

In India, the results reveal a strong, area-driven growth pattern rather than productivity-led expansion. From 1977 to 2023, cocoa area and production exhibited positive growth, while yield showed a declining trend. This imbalance suggests that productivity improvements have not kept pace with the expansion of cultivated land. However, recent years indicate gradual recovery, supported by government programs through the Directorate of Cashew and Cocoa Development (DCCD), technological interventions, and intercropping practices within coconut and arecanut plantations. Kerala, in particular, demonstrates a steady increase in cocoa area and production, confirming its emergence as the leading cocoa-producing state in India.

Time-series and cointegration analyses indicate that domestic and international cocoa prices are interlinked in the long run, reflecting globalization of commodity markets. The price behavior also shows strong seasonal components—prices peaking during lean supply months and declining during harvest periods—suggesting opportunities for improved storage, marketing, and forecasting systems.

Overall, the study concludes that sustainable growth of the cocoa sector depends on enhancing productivity through scientific cultivation, replanting programs, and farmer training, along with ensuring fair and stable prices through policy measures such as the Living Income Differential (LID). Strengthening value addition, promoting export competitiveness, and integrating smallholders into sustainable supply chains are essential for balancing economic, social, and environmental objectives in the global cocoa economy.

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APPENDIX(TABLE)

A)The following table shows area and production of cocoa in kerala from 1980-2024

YEAR	AREA (HA)	PRODUCTION(MT)	PRODUCTIVITY(KG/HA)
1980	1890	243	129
1981	2184	289	156
1982	1960	339	173
1983	2014	412	223
1984	1936	505	261
1985	1920	1542	402
1986	1739	806	463
1987	1663	549	330
1988	1584	549	347
1989	1424	529	371
1990	1418	619	437
1991	1459	945	648
1992	1715	1199	699
1993	1804	1655	917
1994	1894	1492	788
1995	2027	1178	581
1996	2128	685	322
1997	2271	964	424
1998	2820	1445	512
1999	2860	1032	361
2000	3363	1689	502
2001	4000	8500	470.58
2002	4100	8680	472.35
2003	5109	9295	549.65
2004	5870	10220	574.36
2005	5900	10220	577.29
2006	6490	10220	635.02
2007	5800	10220	567.51
2008	6000	10530	569.8
2009	6100	10708	569.66
2010	6344	11044	574.43
2011	7105	11583	613.39
2012	5900	12283	480.33
2013	6136	12483	491.54
2014	6320	13483	468.73
2015	6000	14650	409.55
2016	6500	15344	423.61
2017	7150	15894	449.85
2018	7507	16594	452.39
2019	8507	16894	503.55
2020	9188	16894	543.86

2021	9647	17366	555.51
2022	10130	17770	570.06
2023	10535	18233	577.79
2024	10746	18656	576

SOURCE: DIRECTORATE OF CASHEWNUT AND COCOA DEVELOPMENT (DCCD)
AND DIRECTORATE OF ECONOMICS & STATISTICS (DES)

B) INDIA COCOA AREA ,PRODUCTIONAND YIELD TRENDS FROM 1961 TO
2023

YEAR	AREA(HA)	PRODUCTION(T)	YIELD(KG/HA)
1977	380	300	789.5
1978	630	500	793.7
1979	1000	800	800
1980	20000	6000	300
1981	20000	6000	300
1982	20000	6000	300
1983	20000	6000	300
1984	20000	6000	300
1985	20000	6000	300
1986	20000	6000	300
1987	20000	7000	350
1988	20000	6000	300
1989	20000	6000	300
1990	20000	6000	300
1991	20000	6000	300
1992	20000	6000	300
1993	20000	6000	300
1994	11883	6742	567.4
1995	10727	5772	538.1
1996	11780	6282	533.3
1997	12394	5697	459.7
1998	12038	5281	438.7
1999	12402	5198	419.1
2000	16185	6361	393
2001	15740	6540	415.5
2002	16130	6780	420.3
2003	21893	8397	383.5
2004	25157	9230	366.9
2005	27811	9250	332.6
2006	29471	10175	345.3
2007	30341	10180	335.5
2008	31885	10560	331.2
2009	34049	11820	347.1

2010	46300	12900	278.6
2011	56500	14400	254.9
2012	63000	13000	206.3
2013	66000	13000	197
2014	71000	15000	211.3
2015	78000	16000	205.1
2016	81000	17000	209.9
2017	83000	19000	228.9
2018	89000	20000	224.7
2019	94000	24000	255.3
2020	98000	26000	265.3
2021	103000	27000	262.1
2022	106000	28000	264.2
2023	109000	30000	275.2

SOURCE: FAOSTAT (FOOD AND AGRICULTURE ORGANIZATION OF THE UNITED NATIONS)

C)WORLD COCOA AREA (HA)TRENDS FROM 1961 TO 2023

1961	474270	380000	260000	220000	1756000	6690	700000
1962	464762	380000	277000	220000	1800000	7375	700000
1963	469644	380000	292000	225000	1820000	6659	700000
1964	487136	380000	308000	163000	1848000	7662	700000
1965	482317	380000	327000	240000	1830000	8431	700000
1966	455866	395000	342800	180000	1350000	7783	700000
1967	473078	380000	358000	200000	1450000	9176	700000
1968	432691	370000	371100	210000	1300000	9470	700000
1969	437637	374000	386800	228355	1450000	10041	700000
1970	443916	390000	404300	228262	1451000	8107	700000
1971	447693	396922	422800	219077	1500000	9163	700000
1972	430184	323973	440500	217916	1450000	10607	700000
1973	416175	355692	497000	213106	1400000	8044	700000
1974	508570	337672	471000	221726	1400000	8648	700000
1975	450949	350000	498000	229544	1400000	7376	700000
1976	407329	408371	526000	229500	1300000	8800	700000
1977	412743	431058	698000	238000	1200000	11208	700000
1978	443866	425302	740000	287190	1200000	12553	700000
1979	453569	429161	784700	269994	1200000	16000	700000
1980	482521	444052	836700	269878	1200000	19355	700000
1981	504935	425561	901300	270000	1200000	22500	700000
1982	533273	408009	953000	277000	900000	23000	700000
1983	590744	425000	952600	270000	800000	29000	700000
1984	586242	420000	1028700	265051	800000	43076	700000

1985	649070	426120	1099800	286777	900000	33824	700000
1986	656558	410000	1173700	326405	912000	50632	700000
1987	648476	440000	1233700	310800	739371	61676	700000
1988	667179	420000	1566500	325730	720071	107566	700000
1989	659522	420000	1373300	327980	700286	135020	708000
1990	664853	360000	1566500	328500	693249	158820	715000
1991	667897	350000	1412000	309520	720898	183956	726000
1992	731425	340000	1450000	327060	720898	189596	730000
1993	734124	322521	1425953	320250	702061	299546	735000
1994	698634	350000	1500000	325400	686531	359086	751000
1995	738536	360000	1900000	349370	1000000	374261	788000
1996	661923	367982	2052878	339470	1050000	398655	739000
1997	715671	375301	1823035	341320	1074970	386605	739000
1998	709798	375352	1800000	300119	1364530	403166	743000
1999	681120	370000	1900000	301160	1300000	667897	744500
2000	705965	371325	2000000	402836	1500000	749917	966000
2001	665809	364293	1777550	429547	1350000	765405	966000
2002	582315	365456	1880000	363575	1195000	776901	1030000
2003	590945	450000	2000000	348434	1500000	961107	1002000
2004	638825	490000	2050000	336358	2000000	1090960	1062000
2005	625384	400000	2193548	357707	1850000	1167046	1198902
2006	647135	440000	2281290	350027	1835000	905730	1104000
2007	628928	550000	2372542	356658	1463000	923968	1359550
2008	641337	590000	2300000	376604	1822500	1425216	1349130
2009	635975	600000	2176162	398104	1600000	1587136	1354340
2010	660711	670000	2278786	360025	1600200	1651539	1272430
2011	680484	670000	2685121	399467	1600300	1732600	1241329
2012	684333	670000	2746080	390176	1600300	1852900	1266347
2013	689276	670000	2731411	402434	1600300	1740600	1239750
2014	704122	714261	3084498	372637	1683765	1727400	1144659
2015	702841	767555	3458163	432094	1683765	1709284	1056893
2016	720055	520947	3168749	454257	1683765	1701351	1048945
2017	590813	771917	3982695	467327	1863830	1658421	892101
2018	577191	659017	4072832	501950	1726616	1611014	950132
2019	581897	672618	4308544	525435	1536624	1560945	885571
2020	588501	578736	4309160	527347	1476750	1508955	1188468
2021	600789	608385	4416359	543547	1895400	1421014	1003778
2022	602785	596160	4716003	509179	1219305	1421014	1027173
2023	612775	598804	4791089	516634	1150455	1410871	1045425

SOURCE: FAOSTAT (FOOD AND AGRICULTURE ORGANIZATION OF THE UNITED NATIONS).

D)WORLD COCOA PRODUCTION(MT) TRENDS FROM 1961 TO 2023

Year	Brazil	Cameroon	Côte d'Ivoire	Ecuador	Ghana	Indonesia	Nigeria
1961	155901	75100	85000	44100	415200	1031	197000
1962	140363	76000	110000	44300	428100	897	182000
1963	143495	85000	115000	44300	427800	824	223000
1964	153685	91200	139069	37800	580500	945	298300
1965	160823	78800	121772	49900	415800	1110	184600
1966	172900	86500	149662	50764	381100	1172	267200
1967	194692	93000	146640	61236	423500	1233	238000
1968	149338	102000	144476	81724	327000	1204	191800
1969	211162	108186	180706	47993	414300	1763	220800
1970	197061	133770	179156	53584	406000	1738	304800
1971	218701	138775	225814	70806	470000	2009	256600
1972	221469	126780	185445	67784	415700	1801	241100
1973	195916	132245	208522	63374	343000	1813	215000
1974	242400	120014	241511	91039	385100	3191	214000
1975	281887	96000	231136	75272	397300	3921	216000
1976	231796	82500	232330	65192	326700	3909	181000
1977	249755	107000	303621	72120	277400	4816	193000
1978	284490	110865	318381	72085	268200	5496	157000
1979	336326	123179	397759	77407	280800	8632	151000
1980	319141	117053	417222	91215	277200	10284	153000
1981	335625	118344	464751	80460	246500	13137	174000
1982	351149	105153	360445	96941	202500	17260	156000
1983	380256	108900	411081	45000	168100	19640	140000
1984	329903	120894	565042	77000	166700	26502	160800
1985	430789	118320	555115	130772	194400	33798	160000
1986	459477	123090	610680	89913	226400	34327	148000
1987	328740	132800	664031	57529	188170	50199	150000
1988	374491	129400	832177	85110	246700	79335	253000
1989	392184	125700	780521	82880	295052	110509	256000
1990	256246	115000	807501	96722	293355	142347	244000
1991	320967	105000	764708	90403	241796	174899	268000
1992	328518	97835	813009	93999	312122	207147	292000
1993	340885	99000	803799	82729	254652	258059	306000
1994	330577	107000	808662	81163	288075	241701	323000
1995	296705	134000	1120000	85505	403900	278400	203000
1996	256777	125726	1235300	89928	403000	350800	323000
1997	277966	126807	1119110	90870	322490	329700	318000
1998	280801	125000	1201119	35006	409360	456499	370000
1999	205003	116000	1163025	94687	434200	367475	225000
2000	196788	122600	1401101	64991	436600	421142	338000

2001	185662	122100	1212428	76030	389591	536804	340000
2002	174796	125000	1264708	60268	340562	619192	362000
2003	170004	154965	1351546	88263	497000	695361	385000
2004	196005	166754	1407213	89680	737000	691704	412000
2005	208620	140000	1286330	93658	740000	748827	441000
2006	212270	164553	1408854	87561	734000	769386	485000
2007	201651	212619	1229908	85891	614500	740006	360570
2008	202030	229203	1382441	94300	680781	803593	367020
2009	218487	235500	1223153	120581	710638	809583	363510
2010	235389	264077	1301347	132099	632037.4	844626	399200
2011	248524	240000	1511255	163151.7	700020	712200	391000
2012	253211	268941	1485882	133323	879348	740500	383000
2013	256186	275000	1448992	128446	835466	720900	367000
2014	273793	269228	1637777.56	156216	858720	728400	329870
2015	278299	310000	1796000	180192.4	858720	593331	302066
2016	213871	344752	1634000	177551.3	858720	656817	298029
2017	235809	322247.8	2034000	205955	969300	590684	250000
2018	239318	317530.3	2113189	235182	904700	767280	270000
2019	259451	301255.2	2235043	283679.6	811700	734795	250000
2020	269740	280077	2200000	327902.6	800000	720660	357608.2
2021	302126	295000	2228459	302093.9	1047385	650611.69	280000
2022	301026	290000	2358991	337149.4	683300	650611.69	280000
2023	296145	295819.38	2377442.05	375719.1	653700	641741.02	284232.2

SOURCE: FAOSTAT (FOOD AND AGRICULTURE ORGANIZATION OF THE UNITED NATIONS).

E)WORLD COCOA YIELD(KG/HA) TRENDS FROM 1961 TO 2023

Table 4.8 World Cocoa Yield Trends in Major Producing Countries (1961–2023)

Year	Brazil	Cameroon	Côte d'Ivoire	Ecuador	Ghana	Indonesia	Nigeria
1961	328.7	197.6	326.9	200.5	236.4	154.1	281.4
1962	302	200	397.1	201.4	237.8	121.6	260
1963	305.5	223.7	393.8	196.9	235.1	123.7	318.6
1964	315.5	240	451.5	231.9	314.1	123.3	426.1
1965	333.4	207.4	372.4	207.9	227.2	131.7	263.7
1966	379.3	219	436.6	282	282.3	150.6	381.7
1967	411.5	244.7	409.6	306.2	292.1	134.4	340
1968	345.1	275.7	389.3	389.2	251.5	127.1	274
1969	482.5	289.3	467.2	210.2	285.7	175.6	315.4
1970	443.9	343	443.1	234.7	279.8	214.4	435.4
1971	488.5	349.6	534.1	323.2	313.3	219.3	366.6
1972	514.8	391.3	421	311.1	286.7	169.8	344.4
1973	470.8	371.8	419.6	297.4	245	225.4	307.1
1974	476.6	355.4	512.8	410.6	275.1	369	305.7

1975	625.1	274.3	464.1	327.9	283.8	531.6	308.6
1976	569.1	202	441.7	284.1	251.3	444.2	258.6
1977	605.1	248.2	435	303	231.2	429.7	275.7
1978	640.9	260.7	430.2	251	223.5	437.8	224.3
1979	741.5	287	506.9	286.7	234	539.5	215.7
1980	661.4	263.6	498.7	338	231	531.3	218.6
1981	664.7	278.1	515.6	298	205.4	583.9	248.6
1982	658.5	257.7	378.2	350	225	750.4	222.9
1983	643.7	256.2	431.5	166.7	210.1	677.2	200
1984	562.7	287.8	549.3	290.5	208.4	615.2	229.7
1985	663.7	277.7	504.7	456	216	999.2	228.6
1986	699.8	300.2	520.3	275.5	248.2	678	211.4
1987	506.9	301.8	538.2	185.1	254.5	813.9	214.3
1988	561.3	308.1	531.2	261.3	342.6	737.5	361.4
1989	594.6	299.3	568.4	252.7	421.3	818.5	361.6
1990	385.4	319.4	515.5	294.4	423.2	896.3	341.3
1991	480.6	300	541.6	292.1	335.4	950.8	369.1
1992	449.1	287.8	560.7	287.4	433	1092.6	400
1993	464.3	307	563.7	258.3	362.7	861.5	416.3
1994	473.2	305.7	539.1	249.4	419.6	673.1	430.1
1995	401.7	372.2	589.5	244.7	403.9	743.9	257.6
1996	387.9	341.7	601.7	264.9	383.8	880	437.1
1997	388.4	337.9	613.9	266.2	300	852.8	430.3
1998	395.6	333	667.3	116.6	300	1132.3	498
1999	301	313.5	612.1	314.4	334	550.2	302.2
2000	278.8	330.2	700.6	161.3	291.1	561.6	349.9
2001	278.9	335.2	682.1	177	288.6	701.3	352
2002	300.2	342	672.7	165.8	285	797	351.5
2003	287.7	344.4	675.8	253.3	331.3	723.5	384.2
2004	306.8	340.3	686.4	266.6	368.5	634	387.9
2005	333.6	350	586.4	261.8	400	641.6	367.8
2006	328	374	617.6	250.2	400	849.5	439.3
2007	320.6	386.6	518.4	240.8	420	800.9	265.2
2008	315	388.5	601.1	250.4	373.5	563.8	272
2009	343.5	392.5	562.1	302.9	444.1	510.1	268.4
2010	356.3	394.1	571.1	366.9	395	511.4	313.7
2011	365.2	358.2	562.8	408.4	437.4	411.1	315
2012	370	401.4	541.1	341.7	549.5	399.6	302.4
2013	371.7	410.4	530.5	319.2	522.1	414.2	296
2014	388.8	376.9	531	419.2	510	421.7	288.2
2015	396	403.9	519.4	417	510	347.1	285.8
2016	297	661.8	515.7	390.9	510	386.1	284.1
2017	399.1	417.5	510.7	440.7	520.1	356.2	280.2
2018	414.6	481.8	518.9	468.5	524	476.3	284.2
2019	445.9	447.9	518.7	539.9	528.2	470.7	282.3

2020	458.4	483.9	510.5	621.8	541.7	477.6	300.9
2021	502.9	484.9	504.6	555.8	552.6	457.9	278.9
2022	499.4	486.4	500.2	662.1	560.4	457.9	272.6
2023	483.3	494	496.2	727.2	568.2	454.9	271.9

SOURCE: FAOSTAT (FOOD AND AGRICULTURE ORGANIZATION OF THE UNITED NATIONS).

F) WORLD COCOA AREA ,PRODUCTIONAND YIELD TRENDS FROM 1961 TO 2023

Year	Area(HA)	Production(MT)	Yield(KG/HA)
1961	4403381	1186344	269.4
1962	4468672	1213517	271.6
1963	4515935	1280491	283.5
1964	4544137	1544590	339.9
1965	4591284	1229159	267.7
1966	4076650	1344376	329.8
1967	4220776	1390511	329.4
1968	4079239	1246581	305.6
1969	4260037	1418863	333.1
1970	4358444	1543399	354.1
1971	4423250	1638853	370.5
1972	4299294	1510723	351.4
1973	4316614	1402063	324.8
1974	4398742	1556439	353.8
1975	4360617	1561625	358.1
1976	4309551	1366511	317.1
1977	4436702	1452505	327.4
1978	4581485	1495374	326.4
1979	4632686	1659869	358.3
1980	4740285	1670637	352.4
1981	4848246	1735254	357.9
1982	4678307	1615319	345.3
1983	4658864	1604641	344.4
1984	4768084	1810579	379.7
1985	5045942	2013982	399.1
1986	5247548	2118377	403.7
1987	5274994	2055901	389.7
1988	5655093	2563305	453.3
1989	5514886	2640977	478.9
1990	5721260	2532122	442.6
1991	5654463	2521946.08	446
1992	5726224	2677030.92	467.5
1993	5675475	2672283.72	470.8
1994	5762503	2671268.65	463.6
1995	6556125	2985764.79	455.4

1996	6652960	3237670.67	486.7
1997	6435946	3017083.12	468.8
1998	6672750	3302840.27	495
1999	6862088	2975444.16	433.6
2000	7613598	3338446.71	438.5
2001	7149274	3222354.04	450.7
2002	7013880	3298178.12	470.2
2003	7693381	3705144.75	481.6
2004	8514053	4069753.28	478
2005	8706714	4043867.67	464.5
2006	8514758	4299922.41	505
2007	8644002	3898909.87	451.1
2008	9561187	4265983.94	446.2
2009	9437648	4211454.86	446.2
2010	9592214	4328769.53	451.3
2011	10252360	4553617.54	444.2
2012	10313477	4613754.52	447.4
2013	10177105	4486395.35	440.8
2014	10577940	4742114.16	448.3
2015	10957003	4817236.1	439.6
2016	10699320	4777618.84	446.5
2017	11499069	5235269.43	455.3
2018	11410703	5505866.86	482.5
2019	11351146	5520075.04	486.3
2020	11722785	5606584.09	478.3
2021	11981268	5759894.6	480.7
2022	11656975	5602558.97	480.6
2023	11654929	5600100.42	480.5

SOURCE: FAOSTAT (FOOD AND AGRICULTURE ORGANIZATION OF THE UNITED NATIONS).

G)The following table shows Export and Import of cocoa from 1996-97 to 2024-25

Year	Export		Import	
	Quantity(MT)	Value(Rs. in Crores)	Quantity(MT)	Value(Rs. in Crores)
1996-1997	1252	9.935	4578	37.658
1997-1998	1073	9.646	2937	28.272
1998-1999	933	9.376	4376	42.402
1999-2000	1493	12.985	4367	38.049
2000-2001	1372	13.672	5340	43.272
2001-2002	1361	13.773	6211	50.067
2002-2003	1303	12.556	6483	70.477
2003-2004	2069	19.478	5061	61.212
2004-2005	2446	29.062	9236	94.52
2005-2006	2560	24.802	13509	112.865

2006-2007	3412	40.119	15853	135.057
2007-2008	4979	42.258	19627	189.358
2008-2009	6832	84.039	19084	243.21
2009-2010	5864	96.995	25228	376.078
2010-2011	9078	126.968	33802	584.211
2011-2012	16679	175.976	50760	934.322
2012-2013	19084	293.921	53031	1049.249
2013-2014	16229	573.217	51627	1071.549
2014-2015	20878	848.657	65311	1551.09
2015-2016	32634	1266.994	56425	1398.91
2016-2017	25700	1089.987	63613	1542.307
2017-2018	29583	1144.373	71822	1473.097
2018-2019	27607	1350.86	87595	1845.888
2019-2020	28259	1274.34	85276	1833.974
2020-2021	25768	1108.38	89069	2020.98
2021-2022	27319	1145.48	111187	2713.93
2022-2023	34250	1242.13	128654	3530.75
2023-2024	36186	1521.95	126429	4247.26
2024-2025	18178	2512.4	119935	8426.51

SOURCE : DIRECTORATE OF CASHEWNUT AND COCOA DEVELOPMENT

H)The Following Table Kerala Market Price Of Cocoa From 1999 To 2025

YEAR	PRICE KERALA(RS/QTL)	YEAR	PRICE KERALA(RS/QTL)
1999	1794	2013	3560.00
2000	1375	2014	4720
2001	1298	2015	5033
2002	2172	2016	5045
2003	2316	2017	4018
2004	2181	2018	4058
2005	2026	2019	3987
2006	2012	2020	3954
2007	2088	2021	16208.00
2008	2491	2022	18125.00
2009	3260	2023	22600.00
2010	3960	2024	50041.00
2011	3564	2025	68500
2012	3160.00		

I)The following table shows Domestic price and International price of cocoa from 1994 to 2024

Year	Domestic price (Price Rs./Kg)	International price (Price Rs./Kg)
1994	35.3	37.01
1995	33.4	45.48
1996	48.95	48.4
1997	56.3	49.2
1998	60	55.7
1999	74	65.6
2000	64	38.8
2001	54	44.63
2002	70	78.02
2003	73.71	83.08
2004	74.37	70.32
2005	65.34	69.3
2006	63.58	71.94
2007	67.11	78.58
2008	80	118.65
2009	140	137.05
2010	170	142.74
2011	155	148.9
2012	127	130
2013	126	152
2014	190	215
2015	210	201
2016	225	178
2017	205	134
2018	190	158.45
2019	200	141
2020	175	174
2021	170	181.23
2022	155	186.27
2023	208	228
2024	750	760

SOURCE : DIRECTORATE OF CASHEWNUT AND COCOA DEVELOPMENT

